Cloud Optical Thickness for Satellite-Based Solar Irradiance Estimation and Short-Term Forecasting

Abstract

Cloud optical thickness (COT) describes cloud attenuation of solar radiation, but its added value for irradiance estimation is uncertain when multispectral satellite imagery is already available. We develop a COT-assisted approach that combines local optical-thickness fields retrieved from FY-4B/AGRI imagery with spectral and solar-geometric information to estimate 15-minute mean global horizontal irradiance (GHI). The approach is evaluated against models without COT at three stations in Hubei and two in Hunan, with models trained independently in each region. On the Hubei validation samples, adding COT reduces GHI root mean square error from 74.41 to 68.63 $\mathrm{W\,m^{-2}}$ (7.76%). The largest observed-image reduction, 35.51%, occurs in the thick-cloud subset covered by the satellite reference product and is accompanied by lower positive irradiance bias; improvements under thin and moderately thick clouds are less certain. Applying the irradiance models to predicted imagery extends the approach to 15–240 min forecasting, where COT reduces regional validation RMSE by 4.90% in Hubei and 1.70% in Hunan. Forecast gains depend on cloud regime: thick-cloud subgroup intervals include zero, and moderate-COT conditions in Hubei deteriorate. Input replacements and real cloud sequences indicate useful predicted evolution alongside unresolved errors in optical thickening and thinning. These results support explicit COT as a useful complement to multispectral imagery for irradiance estimation and short-term forecasting, with benefits that vary across cloud conditions and regions.

1 Introduction

Clouds modulate the solar radiation reaching the surface, and changes in their location and optical thickness can produce substantial variability at a ground station. Geostationary multispectral imagery provides frequent observations of these changes and underpins satellite estimates of surface solar irradiance [1]. Cloud optical thickness (COT) describes extinction through a cloud column. Its retrieval from reflected solar radiation has a physical basis, although the inverse problem can be ambiguous for optically thin water clouds [2]. These properties make COT a potentially useful representation for irradiance estimation.

Satellite irradiance methods have long represented cloud attenuation explicitly. Early cloud-index methods relate observed reflectance to clear-surface and cloudy references [3]; Heliosat-2 combines calibrated satellite observations with atmospheric-optics modelling [4]. These indices are distinct from COT. Physical approaches instead use retrieved cloud properties in radiative-transfer calculations or their approximations. Takenaka et al. developed a neural-network approximation of radiative transfer for surface radiation estimation [5], while Heliosat-4 separates clear-sky irradiance and cloud extinction using lookup tables and satellite cloud properties [6]. These studies establish the role of cloud optical properties in radiation estimation; they do not determine how much an explicit COT input adds to a model that already receives multispectral imagery.

Data-driven retrievals learn spectral, spatial, and temporal relationships with surface irradiance. Chen et al. examined spectral selection for global and direct irradiance estimation [7]. STARNet incorporates neighbouring spatial and temporal information from FY-4A [8], and Schuurman and Meyer combined satellite-product emulation with ground-observation fine-tuning for Meteosat irradiance retrieval [9]. Cloud-property information can also be learned separately: Xia et al. retrieved nighttime COT from FY-4B for single-layer non-precipitating clouds [10]. In a hybrid irradiance approach, Liao and Coimbra compared CNNs with and without SCOPE-derived features and extended the approach to short-term forecasting [11]. Together, these approaches motivate testing a cloud-supervised representation alongside the original imagery. In our case, COT is derived from those same observations; it introduces cloud-property supervision without adding an independent measurement.

The contribution of this representation may depend on cloud conditions. Cloud optical properties and aerosols affect radiation errors [12], while satellite viewing geometry and the displacement between a cloud and its surface shadow complicate comparisons with station measurements [13]. Recent FY-4B forecasting work also employs weather-dependent correction of satellite-to-station irradiance differences [14]. Overall error alone therefore gives an incomplete description of performance. We compare the effect of COT within fixed optical-thickness ranges and examine continuous cloud-field and irradiance evolution.

These comparisons link an aggregate gain to the cloud conditions and irradiance variations in which it occurs.

Forecasting introduces uncertainty in the future cloud field. Sirch et al. extrapolated cloud optical thickness for direct-normal irradiance nowcasting [15]; Miller et al. combined cloud advection, cloud properties, and shadow geometry in solar irradiance forecasts [16]. Learned approaches predict different intermediate quantities: IrradianceNet forecasts satellite-derived irradiance fields [17], Chen et al. use a chain of deep-learning models for satellite-based intraday forecasting [18], and Cloud-FNO predicts cloud physical properties [19]. These approaches demonstrate several ways to connect cloud evolution and radiation, with different forecast targets and reference data.

In our image-extrapolation chain, the retrieval and irradiance models receive predicted imagery. Errors in cloud position, boundaries, or growth and decay can reach GHI both through the image input and through retrieved COT. Comparisons of satellite nowcasting methods show that numerical accuracy, spatial appearance, and performance under different sky conditions need not favour the same model [20]. We therefore assess the effect of image prediction through paired input substitutions and aligned cloud and irradiance trajectories, alongside overall error statistics, to examine complementary aspects of forecast performance [21].

We address three questions: (1) how much does local COT improve GHI estimation beyond AGRI imagery and solar geometry; (2) under which optical-thickness conditions and cloud events does the improvement occur; and (3) how much of that benefit remains in 15–240 min forecasts, and how is it constrained by cloud-image extrapolation errors? Hubei provides the primary station comparisons, while independently trained Hunan models provide complementary regional evidence. The work focuses on the contribution of cloud-property representation and its error propagation; developing an improved image-extrapolation model is a subsequent research question.

Table 1: Related approaches and the distinct question evaluated here. This is a comparison of designs, not a cross-dataset ranking of RMSE.

Study	Representation and task	Question informing this study
Chen et al. [7]	Selected multispectral imagery for GHI/DNI estimation	How does a spectral representation support a compact station mapping?
Papachristopoulou et al. [12]	Cloud/aerosol information and radiation modelling for nowcasting and forecasting	How do cloud conditions and forecast lead affect irradiance error?
Zhang et al. [19]	Forecasting cloud physical properties with a Fourier neural operator	Are cloud-property evolution and spatial structure retained?
This study	AGRI retained alongside retrieved COT, with paired representation and forecasting controls	Does explicit COT add GHI value beyond the same imagery, and under which input conditions?

2 Study area and data

2.1 Regions, satellite observations, and cloud reference

The primary region contains Niushou, Wangzhai, and Mizhuang in Hubei. Sili and Zhujia in Hunan form the complementary region. Models and normalization are fitted independently within each region. Regional results are reported separately because station observations have different native temporal sampling.

The AGRI inputs comprise thirteen channels, C01–C06 and C09–C15, with region-specific geolocation. The same station indices register the imagery, geometry, and COT fields. The COT retriever reads a 16×16 patch; the irradiance model selects a station-centred 3×3 neighbourhood from that patch.

The cloud reference consists of archived FY-4B/AGRI cloud physical property (CPP) retrievals, including dimensionless COT and cloud phase, registered to the regional AGRI grid by their latitude and longitude coordinates. Supervised COT pixels require finite reference values in $[0, 100]$ and daytime solar zenith angles below 80°; valid zero-COT values are retained. Hunan additionally requires valid AGRI inputs and a valid clear/water/ice phase category, whereas Hubei uses the COT validity mask indepen-

dently of phase validity. Missing reference values are excluded from COT supervision and reference-based stratification, but CPP availability does not restrict the main GHI comparison. CPP is a satellite retrieval reference, not an independent ground truth; the historical production-version limitations are documented in the supplementary provenance description.

Table 2: Usable station samples for observed-image GHI estimation. Native ground observations are one-minute records in Hubei and five-minute interval means in Hunan. Coordinates are latitude and longitude in degrees; dates are Beijing target dates. Counts are 15-minute station records, with distinct dates in parentheses. Validation counts follow the observation-quality exclusions used for model selection.

Region	Station (latitude, longitude)	Retained target-date range	Training	Validation
Hubei	Niushou (32.0800, 112.0400)	2025-08-01–2026-05-26	6,776 (214)	663 (20)
	Wangzhai (32.0400, 112.1200)	2025-08-01–2026-05-26	6,660 (212)	607 (19)
	Mizhuang (32.1200, 112.2000)	2025-08-01–2026-05-26	6,599 (213)	635 (20)
Hunan	Sili (25.9479, 112.3479)	2024-04-02–2026-06-30	22,930 (508)	1,999 (43)
	Zhujia (29.5667, 112.4500)	2024-04-02–2026-06-30	24,736 (564)	2,227 (50)

The retained GHI samples comprise 20,035 training and 1,905 validation records in Hubei, and 47,666 training and 4,226 validation records in Hunan. Across stations, these cover 214/20 and 578/51 distinct training/validation dates, respectively. Validation target dates span 22 August 2025–4 May 2026 in Hubei and 19 April 2024–25 June 2026 in Hunan; these endpoints do not imply continuous coverage. The Hunan validation set excludes 37 Sili records on 5 September 2024 under the established station-day observation-quality rule. This exclusion is applied during model selection as well as evaluation and removes no training records.

2.2 Irradiance targets and solar geometry

Let τ denote the end of a 15-minute target interval. The target is the interval-mean GHI, $\bar{G}_{s,\tau}$, at station s. Hunan five-minute interval-end records are aggregated as

$$\bar{G}_{s,\tau} = \frac{1}{3}\sum_{j=0}^{2} G^{(5)}_{s,\tau-5j\,\mathrm{min}}. \tag{1}$$

Hubei labels are formed from the valid one-minute bins in the same right-closed interval; incomplete windows are excluded. The clear-sky irradiance $\bar{G}^{cs}_{s,\tau}$ is averaged over that interval using the Ineichen–Perez model [22, 23]. The learning target and reconstruction are

$$k_{s,\tau} = \frac{\bar{G}_{s,\tau}}{\bar{G}^{cs}_{s,\tau}}, \qquad \widehat{G}_{s,\tau} = \widehat{k}_{s,\tau}\bar{G}^{cs}_{s,\tau}. \tag{2}$$

The clear-sky index is not capped at one. GHI observations serve as labels, not model inputs. Solar geometry is computed from UTC, latitude, and longitude and registered with each image. The cosine of solar zenith and the cosine of the solar–viewing relative azimuth provide the geometric inputs; the retriever also receives a daytime indicator.

2.3 Temporal sampling

Training, validation, and final evaluation dates are separated at the date-block level to reduce dependence between fitting and evaluation samples [24]. In both regions, sufficiently covered months reserve separate consecutive two-day blocks for validation and testing, with one-day buffers on each side; months with fewer than twenty eligible days contribute only training data. All frames of a three-frame GHI example or an eight-history/sixteen-future forecasting example must remain in the same partition. Normalization is fitted on training data alone. The fixed imagery partitions contain 34,854/3,171 training/validation sequences in Hubei and 35,320/3,140 in Hunan. These are image-sequence counts, distinct from the station GHI counts in Table 2. Targets and clear-sky quantities are identical within each paired GHI comparison. The results reported here use validation data; the reserved test partitions have not been evaluated.

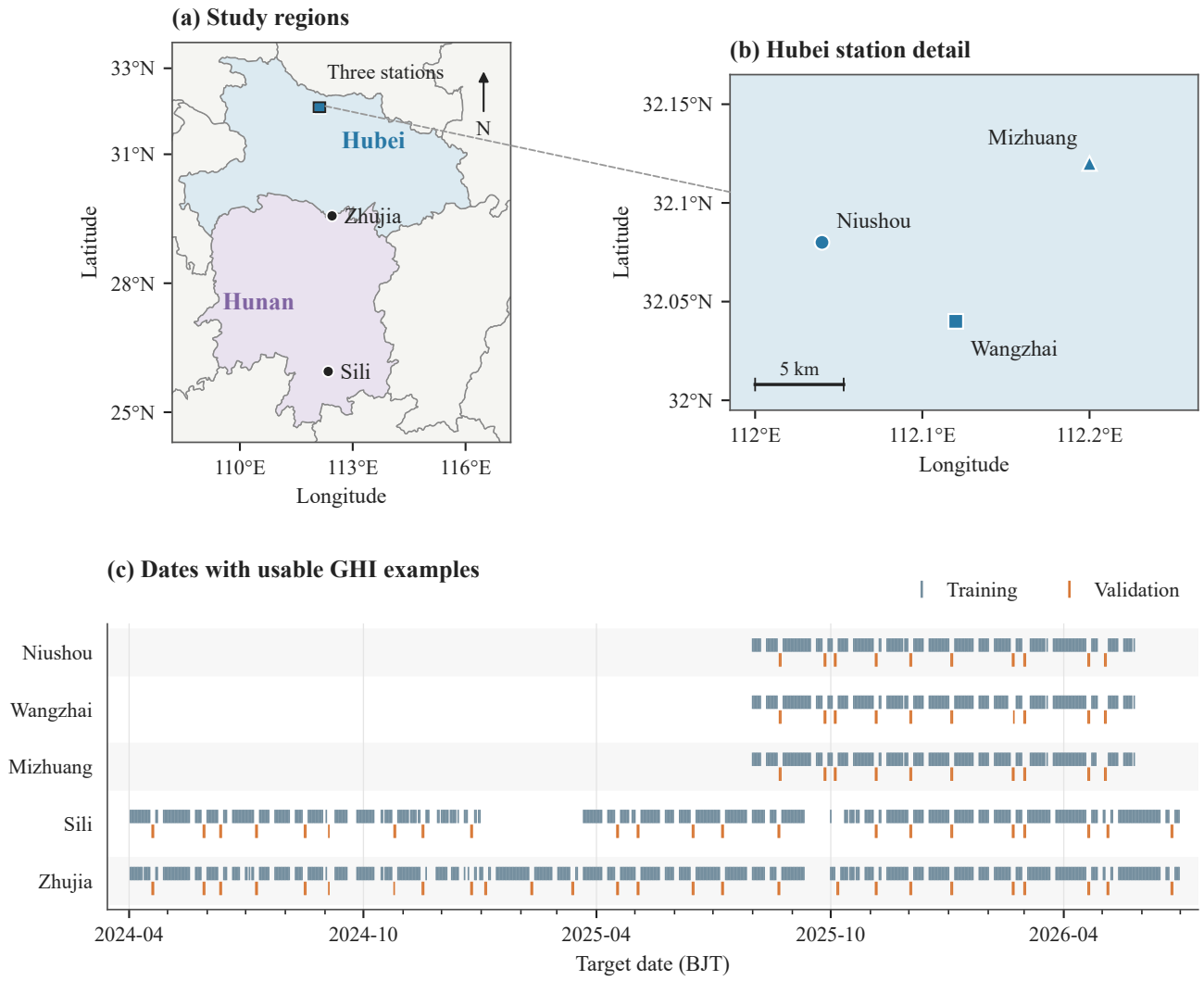


Figure 1: Station locations and usable dates for observed-image GHI estimation. (a) Hubei and Hunan study regions and station locations; (b) local detail of the three Hubei stations. Provincial boundaries are from Natural Earth. In (c), each tick marks a date with at least one valid station sample after the stated quality exclusions; grey-blue and orange ticks distinguish training and validation. Gaps indicate dates absent from the respective usable sample set, rather than zero irradiance. The display concerns paired AGRI–GHI samples, not continuous CPP availability.

3 Methods

3.1 Supervised COT retrieval

For imagery A_τ and geometry g_τ, the retriever produces a nonnegative log-COT field,

$$L_\tau = R_\theta(A_\tau, g_\tau), \qquad C_\tau = \exp(L_\tau) - 1. \tag{3}$$

The retriever follows the U-Net encoder–decoder architecture [25], with two downsampling stages, feature widths 32, 64, and 128 in Hunan and 16, 32, and 64 in Hubei, and skip-connected upsampling to the original 16×16 grid. Convolutional blocks use batch normalization and SiLU; the final softplus produces nonnegative L_τ. It is trained against $\log(1 + C^{CPP})$ with a masked Huber objective [26],

$$\mathcal{L}_R = \frac{\sum_{i,p} m_{i,p} \rho_{0.1}\left(L_{i,p} - \log(1 + C^{CPP}_{i,p})\right)}{\sum_{i,p} m_{i,p}}, \tag{4}$$

where m identifies valid common input and reference pixels. Valid zero-COT references are retained. Training normalization is retained at inference; inference validity uses available inputs and geometry,

without requiring a future CPP reference. Reference-quality criteria apply to supervised training and reference-based evaluation. The retrieval-model parameters remain fixed during irradiance-model training and forecasting.

3.2 Irradiance estimation using local COT fields

Let $\mathcal{P}_s$ select the station-centred 3×3 field. AGRI channels and log-COT are standardized with training statistics:

$$X_{s,\tau,c} = \frac{\mathcal{P}_s A_{\tau,c} - \mu_c}{\sigma_c}, \qquad Z_{s,\tau} = \frac{\mathcal{P}_s L_\tau - \mu_L}{\sigma_L}. \tag{5}$$

The model receives all nine COT values in their original spatial arrangement, preserving local variation and differences in magnitude between samples. The two irradiance models are

$$\widehat{k}^{A}_{s,\tau} = H_A(X_{s,\tau-30:\tau}, g_{s,\tau-30:\tau}), \qquad \widehat{k}^{AC}_{s,\tau} = H_{AC}(X_{s,\tau-30:\tau}, g_{s,\tau-30:\tau}, Z_{s,\tau-30:\tau}). \tag{6}$$

Figure 2 summarizes the estimation model. The primary estimator uses the ordered three-frame sequence at $\tau - 30$, $\tau - 15$ and τ minutes. At each time, the 13 AGRI channels, two geometric channels and standardized log-COT are registered on the same station-centred 3×3 grid. A neutral extra slot gives the implemented tensor shape $[3, 17, 3, 3]$. A temporal convolution maps 17 channels to 32 with a $3 \times 3 \times 3$ kernel; a 3×3 spatial convolution retains 32 channels, followed by ordered flattening and a $288 \to 64 \to 1$ readout. SiLU activations precede the final softplus. Both paired models have 42,529 trainable parameters and receive identical initialization; only the standardized COT slot is set to zero for the baseline. The frame order and local spatial positions are retained, without imposing a cloud-motion trajectory. Original single-frame designs are described in Supplementary material S4.

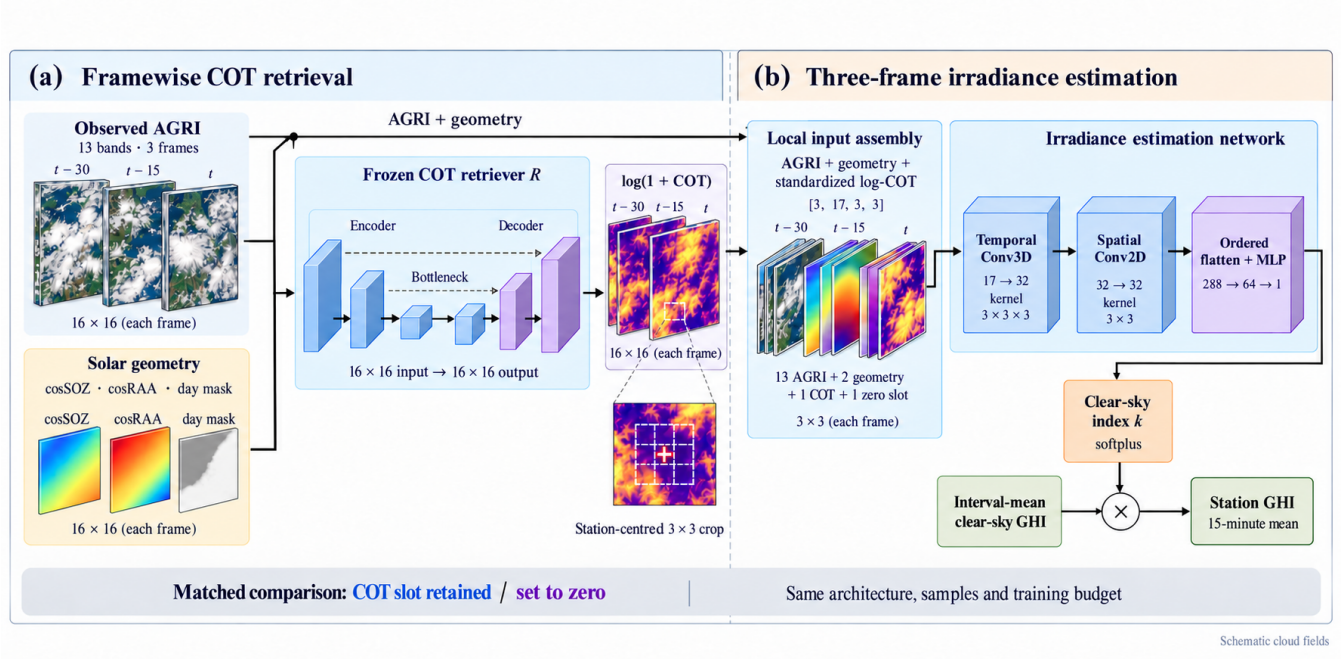


Figure 2: Three-frame COT-assisted GHI estimation. (a) Framewise COT retrieval from observed imagery. (b) Irradiance estimation from the aligned local sequence. Observed AGRI at the target time and 15 and 30 minutes earlier is processed by a frozen retriever. Registered station fields are fused by the implemented temporal and spatial convolutions. The paired models have identical trainable capacity; the standardized COT slot is set to zero in the no-COT model. The extra neutral slot is inactive in both models. AI-generated visual prototype: cloud textures and feature blocks are schematic, rather than measured fields or an exact layer drawing.

The models minimize physical GHI mean squared error, scaled for numerical convenience:

$$\mathcal{L}_H = \frac{1}{N} \sum_i \left[\frac{\widehat{k}_i \bar{G}^{cs}_i - \bar{G}_i}{1000 \text{ W m}^{-2}} \right]^2. \tag{7}$$

Paired training uses the same examples, random seeds, sample order, optimization budget, and selection rule. Table 3 specifies the final architecture and settings for each evaluated configuration.

Table 3: Regional training settings for the models used in the primary three-frame comparisons. Training and validation counts refer to each component's own eligible data; models are trained independently by region. The same settings are used within each with/without-COT pair.

Setting	Hubei	Hunan
COT retriever: 16×16*, 13 AGRI and three geometry/daytime channels*		
U-Net feature widths	16, 32, 64	32, 64, 128
Supervised training/validation patches	2,738 / 264	15,414 / 1,398
Valid training/validation pixels	699,023 / 67,187	3,886,399 / 352,441
Batch; maximum epochs	128; 50	1,284; 50
Input standardization	All training input pixels, all 16 channels	Valid training reference pixels; first 15 channels; binary daytime channel retained
GHI estimator: three 3×3 *frames, 42,529 parameters*		
Batch; epochs	512; 30	1,792; 30
Standardization	Station-specific training statistics	Shared two-station training statistics
Random seeds	42, 43, 44	42, 43, 44
DyDiff image predictor: 256×256*, eight historical and sixteen future frames*		
Training/validation sequences	34,854 / 3,171	35,320 / 3,140
Global batch; epochs	150; 150	80; 150
Updates per epoch	233	442
Sampling	25 steps, one prediction member	25 steps, one prediction member

The COT retrievers use AdamW with learning rate 10^{-3}, weight decay 10^{-4}, gradient-norm clipping at 5, and a maximum of 50 epochs. Training stops after eight epochs without improvement in validation masked log-COT Huber loss; the lowest-loss checkpoint is retained. The GHI estimators use AdamW with learning rate 10^{-3} and weight decay 10^{-4} for 30 epochs, cosine decay to 10^{-5}, and gradient-norm clipping at 5. Each station and random seed selects the checkpoint with the lowest validation GHI RMSE. AGRI and log-COT standardization is fixed from training data, while the two solar-geometric inputs retain their physical values. Initial parameters, sample orders, optimizer settings, and update counts are identical within each COT comparison. The retriever and image-predictor parameters remain fixed when evaluating GHI forecasts.

3.3 Short-term forecasting

Figure 3 illustrates the forecasting application and frame alignment. With initialization t defined by the last historical image, a fixed image predictor S_ψ generates

$$\widehat{A}_{t+k\Delta} = S_\psi(A_{t-7\Delta:t}, g)_k, \qquad \widehat{C}_{t+k\Delta} = \exp R_\theta(\widehat{A}_{t+k\Delta}, g_{t+k\Delta}) - 1, \tag{8}$$

for $k = 1, \ldots, 16$ and $\Delta = 15$ min. SimVP serves as the fixed baseline [27, 28]. For the regional forecasting comparisons, the diffusion-based image predictor DyDiff is trained independently in Hubei and Hunan. Each model is trained for 150 epochs, and its exponential-moving-average weights are selected by validation forecast RMSE. GHI evaluation uses one forecast member generated with 25 sampling steps. Both image predictors use the same downstream comparisons. Future solar geometry is deterministic. Future observed imagery and CPP are excluded from forecasting inputs.

The regional DyDiff predictors share a factorized transformer architecture with patch size 8, width 384, twelve layers, six attention heads, and an MLP expansion ratio of 4 (65,794,266 trainable parameters). They are independently trained for 150 epochs using a 1,000-step mixed-velocity diffusion objective, AdamW with initial learning rate 10^{-4} and weight decay 10^{-4}, two warm-up epochs, a minimum learning rate of 10^{-6}, and gradient-norm clipping at 1. Exponential moving average weights use an update-

dependent warm-up and a maximum decay of 0.999. Model selection uses fixed-seed, single-member forecast RMSE over the full validation set in the training-normalized image space; GHI evaluation uses 25-step single-member sampling. Image normalization is fitted separately on each region's training partition.

We apply the two irradiance models trained on observed imagery to forecast imagery, keeping their parameters fixed, to determine whether the benefit of including COT persists in forecasting. For the three-frame configuration, +15 min includes two known historical frames and +30 min includes one; results for leads ≥ 45 min separately assess inputs consisting entirely of forecast imagery.

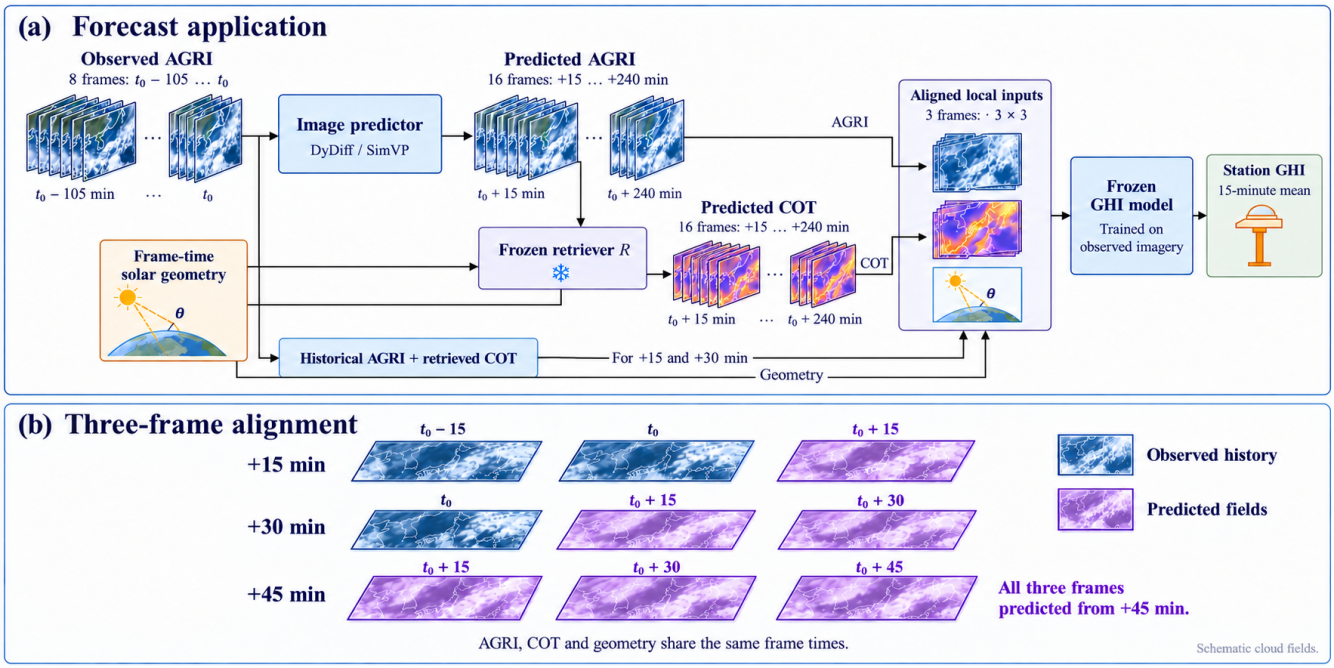


Figure 3: Application of the observed-image-trained irradiance model to forecast imagery. (a) Eight historical AGRI frames enter the image predictor; the resulting sixteen future frames provide both the imagery input and COT retrieved with target-time solar geometry. Registered local fields enter the fixed three-frame GHI model. (b) The three input times for the first three leads: +15 min includes two observed frames, +30 min includes one, and all frames are predicted from +45 min onward. COT is derived from the corresponding observed or predicted frame. The initialization is denoted by t_0 in this diagram; each input uses the geometry at its own time. No GHI model is re-trained on forecast imagery. AI-generated schematic; cloud textures and field tiles are illustrative, not observational or forecast results.

4 Experimental design

4.1 Evaluation of the contribution of COT

Each comparison uses identical station, initialization where applicable, target time, GHI label, and valid daytime mask. Results are computed for each station and random seed before equal-weight averaging within region. We report physical RMSE, MAE, bias, and the relative RMSE reduction

$$\eta = 100\frac{\mathrm{RMSE}_A - \mathrm{RMSE}_{AC}}{\mathrm{RMSE}_A}. \tag{9}$$

Uncertainty is evaluated using 5,000 paired bootstrap draws of target-date blocks defined in Beijing time, retaining overlapping forecast records within the same resampled block. This follows the principle of resampling groups rather than treating correlated records as independent observations [29]. The final tables identify the evaluation partition and date count. Additional image features with matched irradiance-model capacity and features extracted from the fixed COT retrieval model are evaluated to distinguish the effects of model capacity and cloud-property supervision.

4.2 Cloud conditions and continuous events

GHI errors are stratified by station-centre CPP reference COT using fixed near-zero, thin, moderate, and thick bins: $[0, 0.1)$, $[0.1, 5)$, $[5, 30)$, and $[30, 100]$. All models are compared on the same samples with valid CPP references within each bin; the numbers of samples and dates are reported. The overall GHI evaluation also includes samples without CPP observations. Solar-geometry and regional stratification test whether an apparent cloud-thickness pattern reflects other conditions.

Continuous cases combine observed AGRI composites, retrieved and reference COT, and interval-mean GHI. At Niushou, the first station in the fixed study order, candidate validation dates must contain at least twelve of the seventeen 15-minute targets between 10:30 and 14:30 BJT, a station-pixel retrieved COT peak of at least 30, and a COT range of at least 5. We select the earliest eligible date in each of the improving and non-improving RMSE groups. This conditional selection illustrates contrasting responses without selecting the largest changes; the cases do not estimate their frequency. The retrieved-COT condition is separate from the CPP-based thickness strata.

For forecasting, each station–target time is assigned a reference class once, irrespective of how many forecast initializations include it. Consecutive 15-minute thick-reference times are joined; a missing reference, non-thick value, or temporal gap ends the segment. These are station reference segments, not independently tracked cloud objects. Within each region, the earliest multi-time segment in each of the improving and degrading groups is selected. Group assignment uses target-balanced errors over all available origins and three seeds. The displayed initialization maximizes segment coverage, with ties resolved by proximity to a 60-minute advance, without using its forecast error. Displayed curves use seed 42. Image snapshots are taken 30 minutes before the reference peak, at the peak, and at the last forecast target with valid GHI. Future observations are used only for this offline diagnosis. Reference COT at the interval end need not describe the entire 15-minute GHI interval.

4.3 Effects of cloud-image prediction errors on COT and GHI

We keep the retrieval model and the irradiance model with COT unchanged and evaluate all input combinations on the same samples. Let A^o and A^f denote observed and forecast imagery, and let $C^o = R_C(A^o, g)$ and $C^f = R_C(A^f, g)$ denote the corresponding physical COT fields from the same retriever. The four input combinations (A^o, C^o), (A^f, C^o), (A^o, C^f), and (A^f, C^f) isolate sensitivity to the image and COT input routes. The first three use future observations and serve only as offline mechanism experiments.

All comparisons use the same stations, forecast initialization times, target intervals, observations, valid samples, and sample weights. The preprocessing, including log-COT standardization, is identical for every input combination. In the three-frame configuration, known historical positions remain identical; only future positions are replaced. For $a, b \in \{0, 1\}$, define E_{ab} as mean squared GHI error, with 0 denoting observed-derived inputs and 1 forecast-derived inputs. Then

$$E_{11} - E_{00} = \underbrace{(E_{10} - E_{00})}_{D_A} + \underbrace{(E_{01} - E_{00})}_{D_C} + \underbrace{(E_{11} - E_{10} - E_{01} + E_{00})}_{I}. \tag{10}$$

The interaction I allows the GHI response to joint input errors to differ from the sum of separate replacements. This is an exact decomposition of prediction-error sensitivity, not a decomposition into independent physical causes. All terms are evaluated as MSE on the same samples. COT trajectory errors, cloud-boundary displacement, and timing of station irradiance changes provide physical interpretation of the replacement contrasts.

To determine whether image prediction errors reduce the improvement from COT, we also evaluate the irradiance model without COT on observed and forecast imagery using the same samples. Let its MSEs be B_o and B_f. The COT advantages are $V_o = B_o - E_{00}$ and $V_f = B_f - E_{11}$, so

$$V_o - V_f = (E_{11} - E_{00}) - (B_f - B_o). \tag{11}$$

A positive value indicates that replacing observed imagery with forecast imagery reduces the MSE advantage of the model with COT; a negative value indicates that this advantage increases. The sign is evaluated before interpreting any reduction in forecasting benefit.

5 Results

5.1 Evaluation of retrieved COT

On 67,187 valid 16 × 16 CPP-reference pixels from 264 validation frames and 16 target dates, the Hubei retriever has COT RMSE 10.829, MAE 6.054, and bias +1.185 (Fig. 4). These dimensionless measures quantify agreement with the CPP retrieval reference, rather than independent cloud truth. GHI evaluation includes all valid station labels, including dates without CPP; the retrieval-reference and station-GHI sample counts therefore differ.

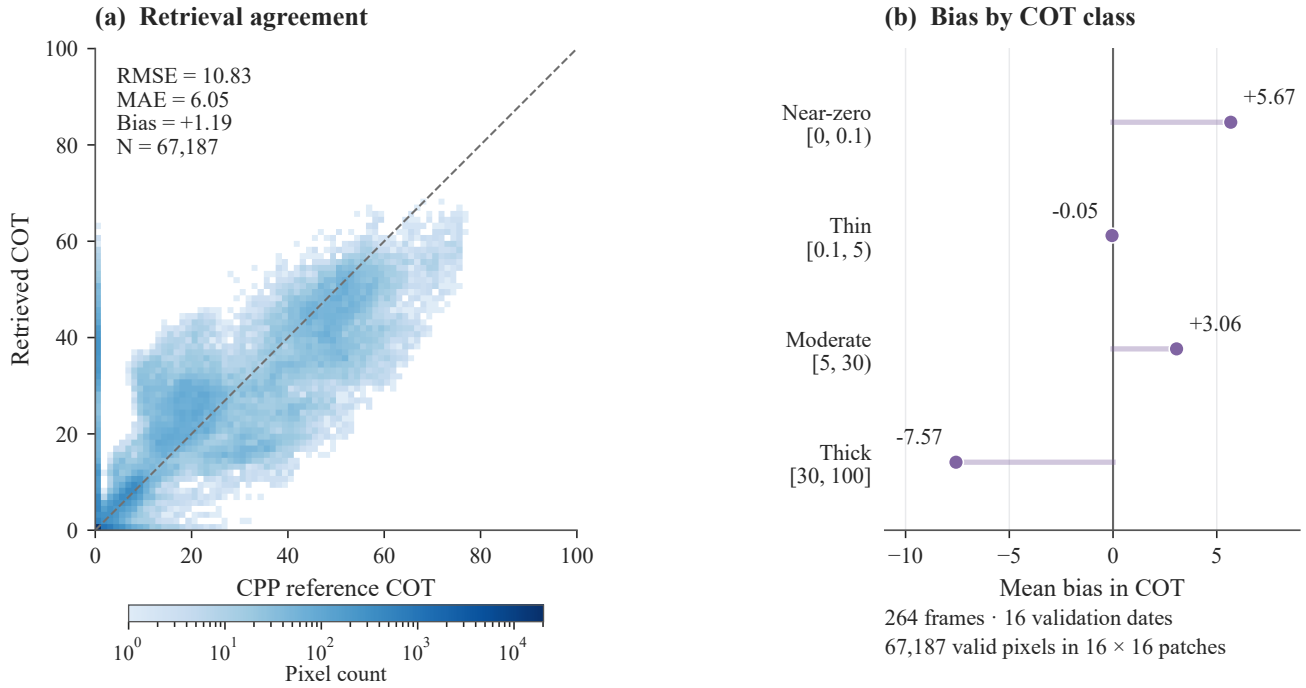


Figure 4: Hubei corrected-geometry COT retrieval: valid-pixel counts against CPP (left) and mean bias within discrete reference thickness classes (right). Scores use all 67,187 valid pixels across 264 frames and 16 target dates. CPP is the retrieval reference. This evaluates the same frozen R used to generate the GHI COT inputs.

5.2 Effect of COT on irradiance estimation

Adding COT to the observed three-frame Hubei inputs reduces station- and seed-averaged GHI RMSE from 74.406 to 68.629 $\mathrm{W\,m^{-2}}$, a 7.76% reduction (Fig. 5). The paired difference is -5.776 $\mathrm{W\,m^{-2}}$ (95% target-date-block interval: $[-9.650, -2.313]$). A consecutive-two-day-block sensitivity analysis gives an interval of $[-10.419, -1.698]$ $\mathrm{W\,m^{-2}}$. All nine station–random-seed pairs improve. The comparison contains 1,905 common station–time records across 20 validation dates; no forecast imagery enters these estimates.

Niushou improves from 69.968 to 64.451 $\mathrm{W\,m^{-2}}$ (7.88%), Wangzhai from 67.452 to 59.215 $\mathrm{W\,m^{-2}}$ (12.21%), and Mizhuang from 85.797 to 82.222 $\mathrm{W\,m^{-2}}$ (4.17%). Mizhuang's date-block interval includes zero, so improvement is not resolved equally at every station. The trainable head capacity, initial weights, example order, optimizer and number of updates are matched within each pair. Thus, additional trainable GHI capacity alone does not explain this paired difference, while the comparison does not isolate COT from every alternative cloud-supervised representation.

The Hubei three-frame data contain 20,035 training records. Target labels are retained from the fixed candidate pool when all three observed frames are available in the same partition. For complementary Hunan evidence, the fixed three-frame models evaluated on the 63,480 forecast-associated validation records give 90.318 $\mathrm{W\,m^{-2}}$ without COT and 89.155 $\mathrm{W\,m^{-2}}$ with COT (1.29%); its paired interval includes zero ($[-3.345, 0.958]$ $\mathrm{W\,m^{-2}}$). These overlapping records use the forecast-comparison weighting and are not 63,480 independent station observations. Original single-frame representation controls remain in Supplementary material S4; their gains are not used as a geometry-only ablation.

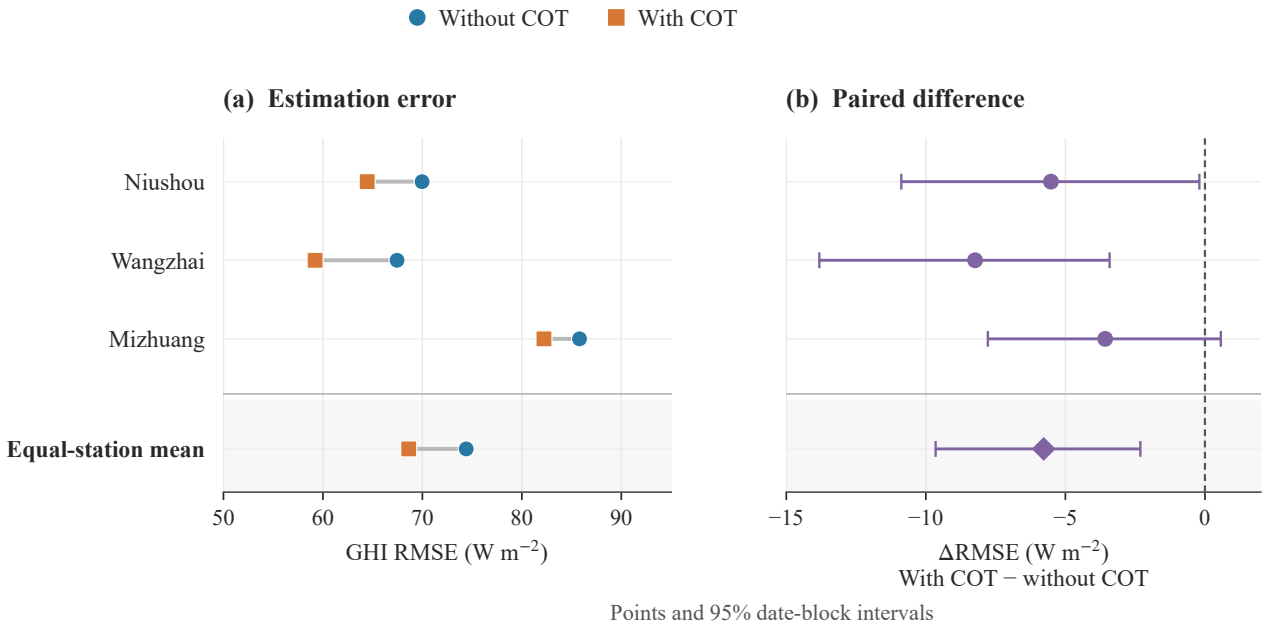


Figure 5: Observed three-frame Hubei GHI estimation. Left: station-specific and equally averaged RMSE, first computed per random seed; grey segments connect the two input groups within each row. Right: paired COT-minus-baseline RMSE differences with 95% target-date-block intervals from 5,000 draws; negative values favour COT. Counts are 663, 607 and 635 validation station–time records at Niushou, Wangzhai and Mizhuang, respectively.

5.3 Dependence on optical thickness and cloud conditions

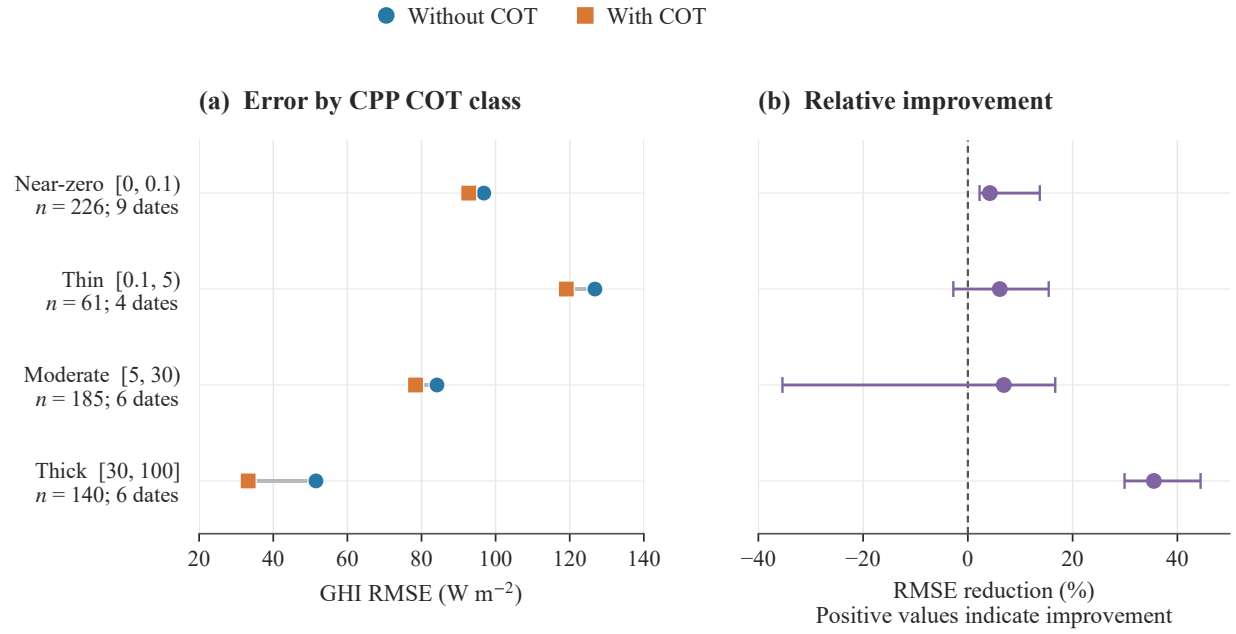
The reference-covered Hubei subsets show RMSE reductions of 4.20%, 6.09%, 6.87%, 35.51% for near-zero, thin, moderate and thick COT, respectively (Fig. 6). These subsets contain 226, 61, 185 and 140 station–time records across 9, 4, 6 and 6 target dates. Thick-cloud RMSE decreases from 51.521 to 33.224 $\mathrm{W\,m^{-2}}$; its relative reduction has a 95% interval of $[29.92, 44.43]\%$. Thick-cloud mean bias decreases from +22.408 to +0.635 $\mathrm{W\,m^{-2}}$, consistent with correcting excessive irradiance under strong optical attenuation.

The thin- and moderate-cloud reduction intervals include zero. Near-zero RMSE decreases, but its positive bias increases from +2.921 to +9.830 $\mathrm{W\,m^{-2}}$; moderate-cloud bias changes from −8.975 to −19.541 $\mathrm{W\,m^{-2}}$. COT therefore improves error magnitude without consistently improving signed bias in every condition. The 612 reference-covered records are a subset of the 1,905 overall GHI records; missing CPP is not interpreted as clear sky. Limited date coverage restricts generalization of the large thick-cloud reduction.

5.4 Cloud events and irradiance response

The two Niushou cases show that COT changes the irradiance response in both directions, rather than imposing a uniform attenuation (Fig. 7). On 26 September 2025, station-pixel retrieved COT ranges from 33.5 to 48.2 while observed GHI remains between 50.4 and 107.1 $\mathrm{W\,m^{-2}}$. Adding COT raises underestimated late-morning values and reduces early-afternoon overestimation. Averaging the three seed-specific RMSEs gives 27.8 $\mathrm{W\,m^{-2}}$ without COT and 19.3 $\mathrm{W\,m^{-2}}$ with COT on fifteen common intervals. Retrieved COT is generally lower than the available CPP reference, yet the added representation still improves GHI in this case.

On 27 September, retrieved COT ranges from 18.4 to 31.7 and is generally higher than the CPP reference. At 14:00, observed interval-mean GHI rises to 484.2 $\mathrm{W\,m^{-2}}$ while both models remain much lower. The COT-assisted model gives the lower prediction and increases RMSE from 92.7 to 100.9 $\mathrm{W\,m^{-2}}$ over sixteen common intervals. The juxtaposition of the cloud field, station COT and GHI shows that a local thickness representation can improve a low-irradiance period but does not guarantee that rapid irradiance changes are captured. These adjacent-date examples are not independent repetitions and do not isolate retrieval error from spatial or temporal representativeness.



n: station–time records; intervals: 95% target-date-block bootstrap.

Figure 6: Hubei GHI errors (left) and paired relative reductions (right) conditional on valid station-centre CPP reference COT. Grey segments connect errors with and without COT within each class; positive reductions favour COT. Horizontal intervals use 5,000 paired target-date draws, retaining fixed station–seed groups. Labels give common station–time records and target dates; records without CPP remain in the overall GHI evaluation.

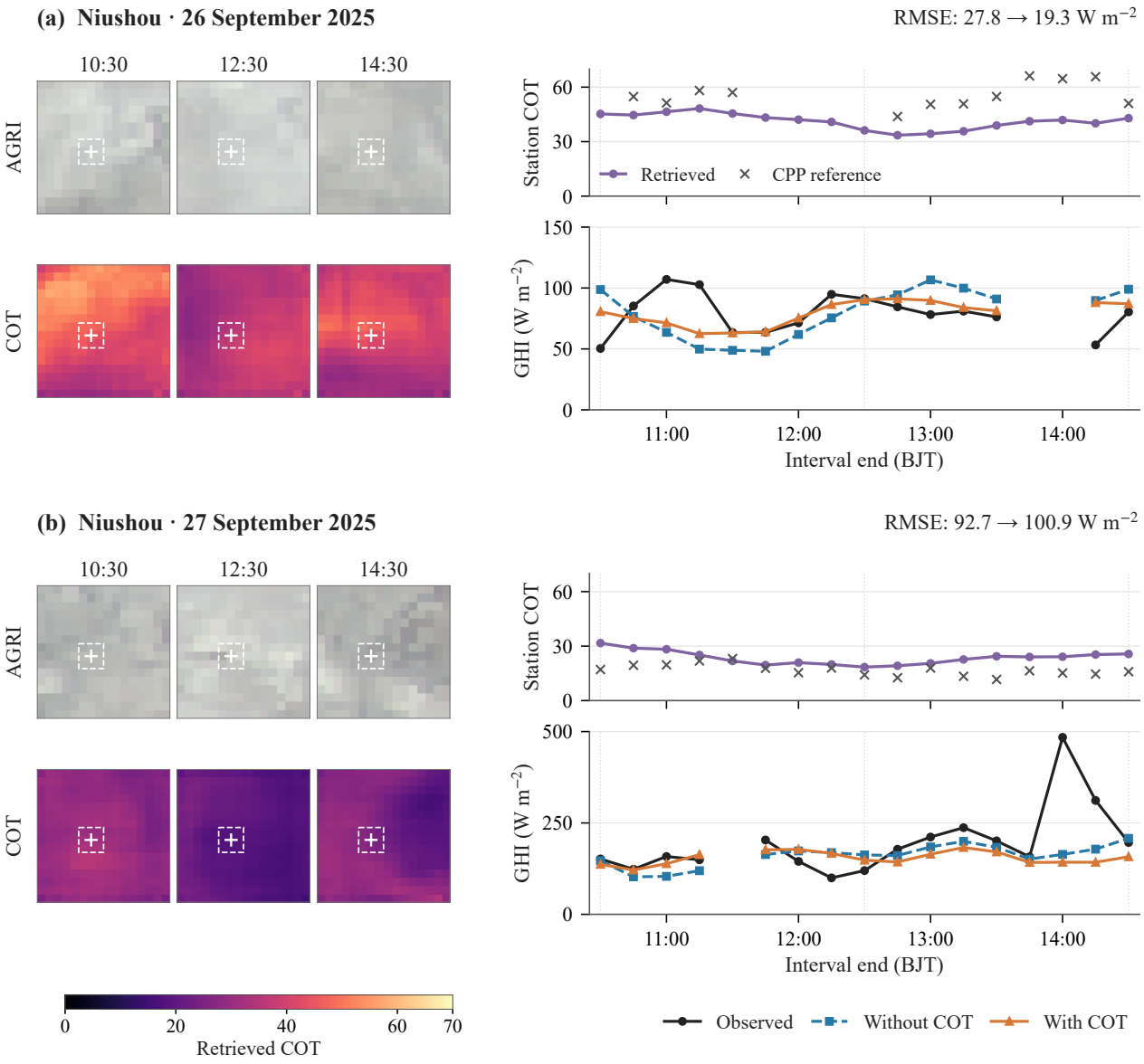


Figure 7: Observed-image GHI estimation at Niushou on (a) 26 and (b) 27 September 2025 using the current three-frame models. Each row shows observed AGRI composites and retrieved COT at 10:30, 12:30 and 14:30 BJT on the left, with station-pixel COT and GHI trajectories on the right; vertical dotted guides mark snapshot times. Each snapshot is the last frame of the corresponding three-frame input. Composites use C03/C02/C01 with a common square-root transform of reflectance clipped to [0,1]; the native 16×16 pixels are retained. Crosses and dashed boxes mark the station and its 3×3 input. COT maps and curves share scales; GHI limits differ (0–150 and 0–500 W m^{-2}). Sparse CPP references are unconnected crosses. GHI observations are target-ending 15-minute means; displayed model curves average three seeds, while RMSE is calculated per seed and then averaged. Missing targets remain gaps. Selection follows Section 4.2 and represents contrasting cases, not their prevalence.

5.5 Effect of COT on forecasting and propagation of image prediction errors

Adding predicted COT improves the overall GHI forecast in both independently trained regions when the observed-image-trained irradiance models are applied without parameter updates to DyDiff imagery (Table 4). In Hubei, station- and seed-averaged RMSE decreases from 101.018 to 96.071 $\mathrm{W\,m^{-2}}$ (4.90%). The paired difference is -4.947 $\mathrm{W\,m^{-2}}$, with a 95% target-date-block interval of $[-9.983, -0.154]$. The evaluation includes all 882 forecast windows with eligible GHI labels within the common validation dates of the image and irradiance models: 24,451 station–initialization–target records across 17 target dates, corresponding to 1,566 distinct station–target observations. No selection is made by weather, CPP availability, or forecast error. All nine station–random-seed pairs improve; the station-specific reductions range from 4.69% to 5.28%, although each station-specific date-block interval includes zero.

In Hunan, RMSE decreases from 122.588 to 120.502 $\mathrm{W\,m^{-2}}$ (1.70%) on 66,086 records across 51 target dates, with a paired difference interval of $[-3.527, -0.598]$ $\mathrm{W\,m^{-2}}$. Sili improves by 2.47% and Zhujia by 0.76%; the Zhujia-specific interval includes zero. The two regional comparisons therefore show overall forecast improvements from the added COT representation, while the magnitude and uncertainty vary by station. Regional observation periods and samples differ, so absolute RMSEs are not a ranking of regional model quality.

Table 4: GHI forecasting with DyDiff imagery, with and without predicted COT, over +15–240 min. Both inputs include solar geometry. Irradiance models are trained on observed three-frame sequences and held fixed during forecasting. RMSE ($\mathrm{W\,m^{-2}}$) is computed separately for each seed before averaging; regional means weight stations equally. Hubei and Hunan use 24,451 records from 17 validation target dates and 66,086 records from 51 dates, respectively.

Region	Station	No COT	Predicted COT	Reduction (%)
Hubei	Niushou	93.890	89.431	4.75
	Wangzhai	96.595	91.490	5.28
	Mizhuang	112.569	107.292	4.69
	Equal-station mean	**101.018**	**96.071**	**4.90**
Hunan	Sili	134.723	131.395	2.47
	Zhujia	110.453	109.609	0.76
	Equal-station mean	**122.588**	**120.502**	**1.70**

At +45–240 min, all three input frames are predicted. Hubei RMSE decreases from 104.197 to 99.273 $\mathrm{W\,m^{-2}}$ (4.73%) on 21,324 records; the paired interval of $[-10.124, 0.093]$ $\mathrm{W\,m^{-2}}$ narrowly includes zero. Hunan RMSE decreases from 126.172 to 124.012 $\mathrm{W\,m^{-2}}$ (1.71%) on 57,907 records, with an interval of $[-3.632, -0.638]$ $\mathrm{W\,m^{-2}}$. Thus the point improvements are not confined to the first two leads containing observed historical frames, although the all-predicted Hubei subset has greater uncertainty.

The point estimates favour COT at all sixteen leads in both regions (Fig. 8). At +60 and +240 min, relative reductions are 6.07% and 2.06% in Hubei, and 2.09% and 1.19% in Hunan. Some lead-specific date-block intervals include zero, particularly at longer horizons in Hubei; the overall improvements therefore do not establish an individually resolved benefit at every lead.

To compare image predictors and diagnose input errors, the following Hunan analyses retain the original common cohort of 63,480 records across 51 target dates. On that cohort, DyDiff RMSE decreases from 124.958 to 122.788 $\mathrm{W\,m^{-2}}$ (1.74%) with COT, with a paired difference interval of $[-3.685, -0.735]$ $\mathrm{W\,m^{-2}}$. Using the same samples and irradiance models with SimVP imagery gives 127.421 versus 125.340 $\mathrm{W\,m^{-2}}$ (1.63%). The image-predictor contrasts are retained in Fig. S9. Although DyDiff has lower point RMSEs, paired intervals for its difference from SimVP include zero for both irradiance models; this comparison does not establish superiority of one image predictor.

A controlled COT replacement indicates that errors in predicted COT still limit the accuracy of the COT-assisted forecast. Keeping the DyDiff AGRI input and the COT-assisted irradiance model fixed, replacing predicted COT at future positions with COT retrieved from observed future imagery decreases RMSE from 122.788 to 108.941 $\mathrm{W\,m^{-2}}$; the paired difference is -13.847 $\mathrm{W\,m^{-2}}$ (95% interval: $[-17.582, -10.636]$). Conversely, replacing future COT with its last observed value increases RMSE to

129.516 W m^{-2}, a paired increase of 6.728 W m^{-2} ([3.587, 10.035]). The former is an offline observed-input diagnostic, not an attainable forecast score; together, the contrasts indicate that predicted COT carries useful evolution information while retaining substantial error.

The current regional three-frame evaluation does not include the complete four-way substitution in Eq. 10; the reported COT-source contrasts are therefore input sensitivities rather than a quantitative allocation of error among the image route, COT route, and their interaction.

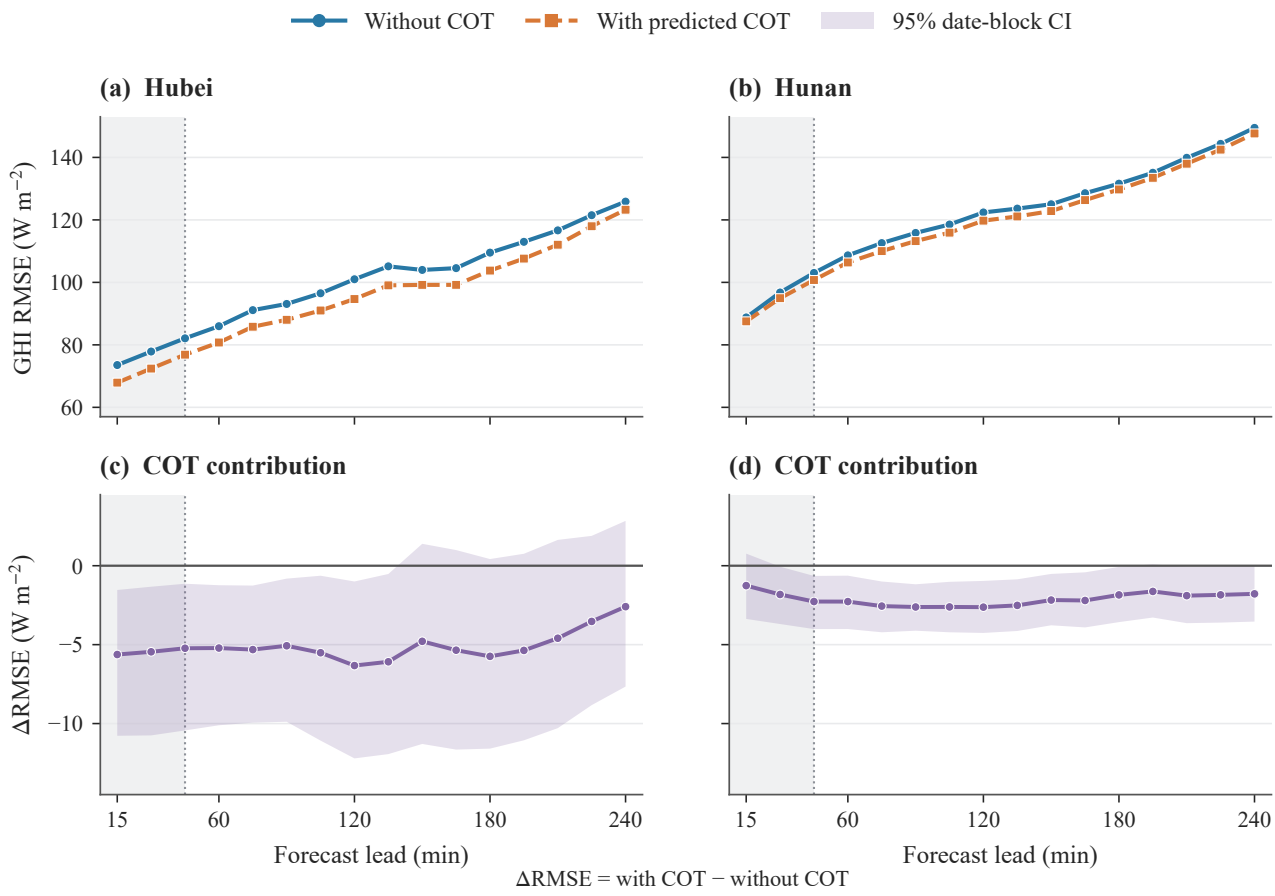


Figure 8: Lead-dependent GHI forecast errors for the independently trained regional DyDiff predictors and fixed observed-image-trained irradiance models. (a,b) RMSE without and with predicted COT; (c,d) their paired difference (with minus without COT), with pointwise 95% intervals from 5,000 target-date-block bootstrap draws. Scores average station- and seed-specific RMSEs equally. Hubei uses 24,451 records from 17 dates and Hunan 66,086 records from 51 dates; counts vary by lead. Grey shading marks the first two leads, which include known historical imagery; all three frames are predicted from +45 min onward. Both models include solar geometry.

5.6 Cloud-regime dependence of forecasting

The overall forecast improvement does not extend uniformly to every reference thickness class (Fig. 9). Under thick reference COT, Hubei RMSE decreases from 72.479 to 61.281 W m^{-2} (15.45%), and Hunan from 92.134 to 89.464 W m^{-2} (2.90%). The paired target-date-block intervals are $[-25.927, +1.089]$ and $[-5.821, +0.690]$ W m^{-2}, respectively; both include zero. These subsets contain 139 distinct station–target observations on six dates in Hubei and 165 on 25 dates in Hunan. In contrast, Hubei moderate-COT RMSE increases from 86.965 to 94.089 W m^{-2}, with a difference interval of $[+4.320, +24.961]$ W m^{-2}.

Thick-cloud mean GHI bias decreases from +31.986 to +16.072 W m^{-2} in Hubei and from +34.456 to +22.821 W m^{-2} in Hunan. Thus the point improvement is associated with reduced positive bias, but the available reference dates do not resolve a stable thick-cloud forecast benefit. Only 32.6% and 28.4% of the regional forecast records have a quality-valid station CPP reference. The stratified results describe this covered subset; they neither replace the full-cohort forecast scores nor isolate the loss of benefit relative to observed-image estimation.

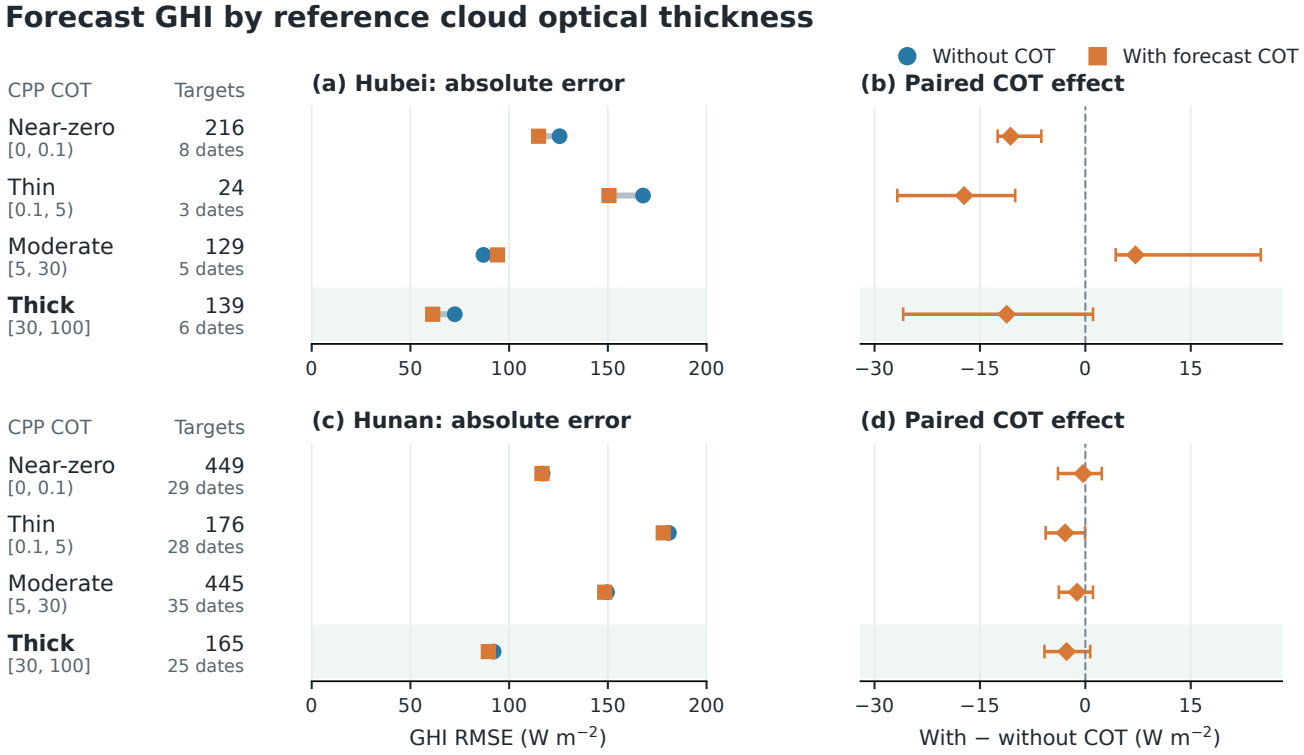


Figure 9: GHI forecast errors conditional on target-time CPP optical thickness in Hubei and Hunan. Left: RMSE without and with predicted COT. Right: the paired difference (with minus without COT) and 95% intervals from 5,000 target-date-block draws; negative values favour COT. Scores first average the three seed-specific RMSEs within each station and then weight stations equally. Repeated origins are retained in scoring; labels count distinct station–target observations and dates. Classes are $[0, 0.1)$, $[0.1, 5)$, $[5, 30)$, and $[30, 100]$. CPP is an offline retrieval reference, and near-zero does not by itself establish clear sky.

The reference merging yields 75 Hubei and 137 Hunan station segments, but only 18 and seven, respectively, contain at least two consecutive thick-reference times. This fragmentation limits process-level inference. Figure 10 contrasts one improving and one degrading multi-time segment, while all four selected cases and exact coverage are retained in Supplementary material S7. At Mizhuang on 26 September 2025, the forecast COT weakens before the sustained thick reference does, yet including it reduces part of the GHI overestimation. For the displayed initialization and seed, thick-segment RMSE decreases from 26.67 to 14.51 $\mathrm{W\,m^{-2}}$ over nine targets.

At Sili on 15 June 2025, the forecast COT misses renewed local thickening and later remains elevated as the observed cloud becomes optically thinner. The displayed thick-segment GHI RMSE increases from 14.87 to 22.10 $\mathrm{W\,m^{-2}}$ over three targets. Later in the forecast, both GHI models miss much of the observed irradiance recovery. The juxtaposition shows that a useful aggregate COT contribution can coexist with imperfect cloud evolution and adverse event corrections. The examples describe conditional responses; they do not uniquely separate image-prediction, retrieval, or station-mapping errors.

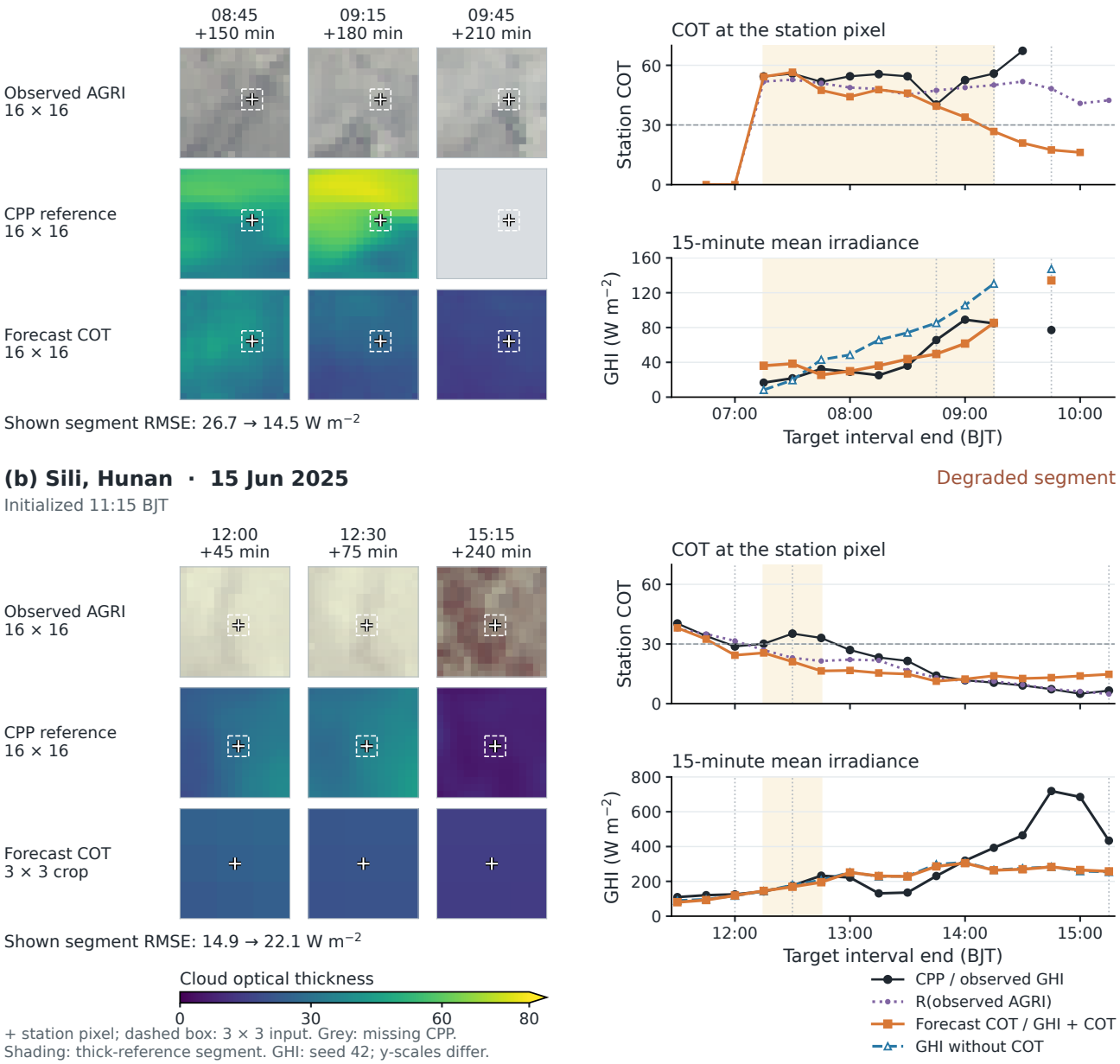


Figure 10: Contrasting forecast cases: (a) Mizhuang, 26 September 2025, initialized at 06:15 BJT; (b) Sili, 15 June 2025, initialized at 11:15 BJT. Observed AGRI, CPP and predicted COT snapshots accompany station-COT and GHI trajectories. Composites use C03/C02/C01 and a common square-root transform of reflectance clipped to [0,1]. Native pixels are retained: Hubei fields are 16×16; the archived Hunan forecast COT is a magnified 3×3 crop, not the full observed field of view. Crosses mark the station; boxes locate its 3×3 neighbourhood. COT maps share a scale, while GHI axis limits differ. Grey maps and curve gaps indicate missing values. Shading marks the selected thick-reference segment, and dotted lines mark snapshots. GHI curves use one initialization and seed 42; observations are target-ending 15-minute means. Future observed inputs appear only as offline references. Cases follow the outcome-conditional selection in Section 4.2.

6 Discussion

6.1 Contribution of COT to irradiance estimation

COT describes optical attenuation by clouds and provides a physically interpretable input to the irradiance model. Training the retrieval model against cloud products may help the irradiance model use cloud information that is implicit in the spectral inputs. Comparisons with additional image features and retrieval-model features assess how much of the improvement is associated with increased model capacity or cloud-property supervision. The original-geometry comparisons in Supplementary material S4 suggest a contribution from cloud-product supervision and motivate evaluating thickness-dependent bias. Their absolute gains do not establish the corresponding effects for a model using different solar geometry. The Hubei three-frame result provides a matched-head-capacity demonstration of a COT-related GHI improvement. The larger thick-cloud correction is associated with lower positive irradiance bias; the smaller Hunan point reduction is not resolved from zero. Cloud-supervised representation is therefore useful under some evaluated conditions, rather than uniformly beneficial across regions and cloud regimes.

Thickness-dependent gains should be interpreted through both attenuation and irradiance variability. Radiative-transfer-based estimation links optical-property errors to surface fluxes, while unresolved broken clouds complicate agreement with point observations [5]. Cloud edges, diffuse radiation, and changes during a 15-minute interval further complicate the mapping to station GHI. Cloud-shadow displacement and viewing parallax can also change which cloud pixels are relevant to a station [16, 13]. The continuous cases illustrate contrasting irradiance responses to the added representation, but do not separate these physical processes.

6.2 Cloud-image extrapolation as a constraint on forecasting benefit

The regional forecast results extend the evidence for COT from observed-image estimation to predicted-image inputs: overall RMSE decreases by 4.90% in Hubei and 1.70% in Hunan without adapting the irradiance models to forecasts. These are regional validation comparisons, not evidence of uniform gains across stations or cloud conditions. The Hubei forecast reduction is numerically smaller than the 7.76% observed-input reduction, but their samples and temporal weighting differ. That difference alone cannot quantify the effect of image extrapolation on the COT benefit.

The cloud-regime analysis adds a qualification to this aggregate benefit. Thick-cloud positive bias decreases, but both regional thick-subset intervals include zero, and Hubei moderate-COT errors increase. The Mizhuang example improves despite errors in predicted optical thickness, whereas the Sili example shows that adding COT does not recover the missing irradiance rise. COT accuracy and incremental GHI value are therefore related but not interchangeable: their connection depends on the joint spectral inputs and the irradiance mapping.

Errors in predicted cloud images can affect GHI directly through the spectral and spatial inputs and indirectly through the COT retrieved from those images. Errors in cloud position can change the location and timing of attenuation at the station. Spatial smoothing can reduce local COT contrasts, while errors in cloud growth and dissipation can suppress or delay changes in optical thickness. Prior nowcasting studies distinguish spatial appearance and numerical errors, and identify difficulties in representing cloud evolution [18, 20]. If the COT-related improvement is smaller with forecast imagery, these processes could help explain the reduction.

The input-replacement results distinguish two questions: whether COT helps relative to an imagery baseline, and how accurate the COT-assisted model could be with better COT inputs. On the same Hunan records and with the same three-frame models, observed imagery gives 90.318 $\mathrm{W\,m^{-2}}$ without COT and 89.155 $\mathrm{W\,m^{-2}}$ with COT, a 1.29% reduction whose paired interval includes zero ($[-3.345, 0.958]$ $\mathrm{W\,m^{-2}}$). This point reduction is smaller than the 1.74% obtained with DyDiff imagery. Hence, these comparisons do not show that extrapolation reduces the relative benefit of COT. Nevertheless, the large improvement after replacing forecast-derived COT with observed-derived COT indicates a sensitivity to errors entering through this input. It does not separate cloud-image prediction error from retrieval behaviour on predicted images or the irradiance model's response to the substituted inputs. The single-image and three-frame percentage gains also involve different samples and representations and cannot be interpreted as an error

budget.

Improving the prediction of cloud position, boundaries, and optical-thickness evolution is therefore a direction for further research. Holding the retrieval and irradiance models fixed would allow future work to test whether more accurate cloud-image predictions lead to further reductions in GHI forecast error.

6.3 Applicability and limitations

The applicability of the results depends on regional sampling, reference-product uncertainty, and the number of independent weather periods evaluated. CPP-based error measures assess agreement with a retrieval reference, whereas GHI performance is assessed against station observations. Ground-reference quality also matters: a recent FY-4B validation study showed that radiometer quality can materially affect the assessment of satellite irradiance products [30]. Representation controls and adverse cases remain relevant to the interpretation even when the average COT effect is positive. The analysis uses development-validation dates, with the reserved test partitions unevaluated. Sparse CPP coverage limits cloud-regime and continuous-event conclusions: most reference segments contain a single time, and endpoint COT is not an interval-integrated cloud measurement. The cases and input sensitivities constrain possible explanations but do not attribute errors uniquely to cloud motion, optical evolution, retrieval, or radiative effects.

7 Conclusions

This study evaluates explicit COT as a local representation for satellite GHI estimation and follows its effect into short-term forecasting. In Hubei, COT reduces observed three-frame GHI RMSE by 7.76%, with point improvements at all three stations. The reference-covered thick-cloud subset shows a 35.51% reduction and substantially lower positive bias, while thin- and moderate-cloud intervals include zero. The complementary Hunan observed-input comparison gives a smaller 1.29% point reduction with an interval crossing zero. With independently trained regional DyDiff predictors and fixed observed-trained three-frame irradiance models, 15–240 min validation RMSE decreases by 4.90% in Hubei and 1.70% in Hunan. At +45–240 min, when all three frames are predicted, the point reductions are 4.73% and 1.71%, respectively; the Hubei subset interval includes zero. These results extend the observed-input evidence to an overall forecasting benefit in both regions, while unresolved thick-cloud subgroup intervals and moderate-COT degradation in Hubei demonstrate its cloud-regime dependence. Fixed-model COT replacements indicate both useful predicted evolution and remaining sensitivity to COT input errors; they do not identify a unique physical cause or demonstrate a reduction of relative COT benefit due to image extrapolation. The resulting evidence provides a basis for improving irradiance-relevant cloud evolution in future image-extrapolation models.

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Supplementary material

S1. Model sizes and comparison inputs

The archived CPP files provide COT, cloud phase, and geolocation, but the recovered processing records do not uniquely identify a public operational product version or a uniform producer quality flag. We therefore describe them as archived FY-4B/AGRI retrieval references, rather than assigning an unverified NSMC product identity. The numerical and input-validity conditions in Section 2.1 are the actual filters used here. Regional reference availability and validity filters differ; neither COT-reference errors nor reference-covered subsets constitute independent ground-truth validation.

For reproducibility, the Hubei retriever standardizes its inputs using all 3,178 training input patches, including patches without valid CPP supervision; the 2,738 supervised patches in Table 3 are a subset. The Hunan GHI models retain a fixed training-only log-COT scale shared across the two stations and pooled retrieval calibration outputs, with mean 1.8354828989 and standard deviation 1.3974485910. These statistics are not re-fitted on validation data or forecast imagery.

The parameter counts in Table S1 describe the implemented model classes, excluding optimizer states. They do not by themselves establish equivalent regional training or predictive performance. The original single-image Hunan auxiliary-image and retrieval-feature variants match one another but each contains 124 more trainable head parameters than the explicit-COT variant. Those original single-frame counts do not describe the three-frame heads, which are listed separately in the first row. The Hubei four-arm synchronous design uses identical tensor shapes and parameter counts with inactive auxiliary slots set to zero.

Table S1: Implemented model dimensions and parameter counts. A retrieval model is frozen while fitting a GHI estimator. Head counts include the trainable station mapping; they do not include the separate frozen retriever.

Region and component	Input / feature definition	Readout width	Parameters
Both regions, three-frame GHI	Temporal $17 \to 32$, spatial $32 \to 32$; 3×3 fields	288	42,529
Hunan COT retriever	16×16, 16 channels; widths 32–64–128	—	470,913
Hubei COT retriever	16×16, 16 channels; widths 16–32–64	—	119,233
Hunan imagery head	3×3 AGRI + geometry; $32 \times 3 \times 3$ features	288	22,177
Hunan COT head	Imagery features + four COT feature channels	324	24,489
Hunan auxiliary-image head	Additional learned image features	324	24,613
Hunan retrieval-feature head	Hidden features of the frozen retriever	324	24,613
Hubei original single-frame head	Two auxiliary slots mapped $2 \to 4$; inactive slots zero	324	24,489

Table S2: Inputs and fitted components in the four-arm forecasting comparison. All arms receive the same forecast AGRI and deterministic solar geometry. Historical observations end at initialization; forecast COT is derived from predicted AGRI.

Comparison	Historical COT	Future COT	Fitted component
No COT	Neutral-zero slots	Absent	Base irradiance model
Historical COT	Eight observed-history retrieved fields	Absent	Otherwise matched base model
Historical COT + control	Same frozen historical-COT base	Neutral-zero slots in the correction	Gated signed residual head
Historical COT + forecast COT	Same frozen historical-COT base	Forecast-image COT and changes	Equal-capacity gated signed residual head

Base and correction stages each use 30 epochs, batch 1,536, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, and cosine decay to 10^{-5}; seeds are 42, 43 and 44. Base and correction dates are disjoint. Correction outputs are zero-initialized, signed, and combined with a sigmoid gate. Both correction arms use the same future-input availability mask; unavailable input returns the frozen base prediction. This table concerns the forecasting experiment, not the Hubei synchronous auxiliary-slot comparison.

S2. Conditional COT probe on development dates

A separate exploratory probe uses 32,838 common records from 50 development target dates between 6 August 2025 and 12 July 2026. A deterministic ridge correction is evaluated with three target-date folds; each of three stations is assessed using three frozen baseline sources. These sources are not three newly trained probe seeds. Mean GHI RMSE is 118.799 W m^{-2} for the existing frozen baseline, 124.850 for context-only correction, 124.307 after adding forecast COT to that context, and 99.569 when actual future imagery supplies the retrieved COT instead. Thus the forecast-COT probe remains worse than the existing baseline even though it slightly improves the context-only probe. The observed-future alternative is an offline information diagnostic. These development rows differ from the validation cohort in Fig. S7; their absolute RMSE differences are not a validation-to-oracle gain.

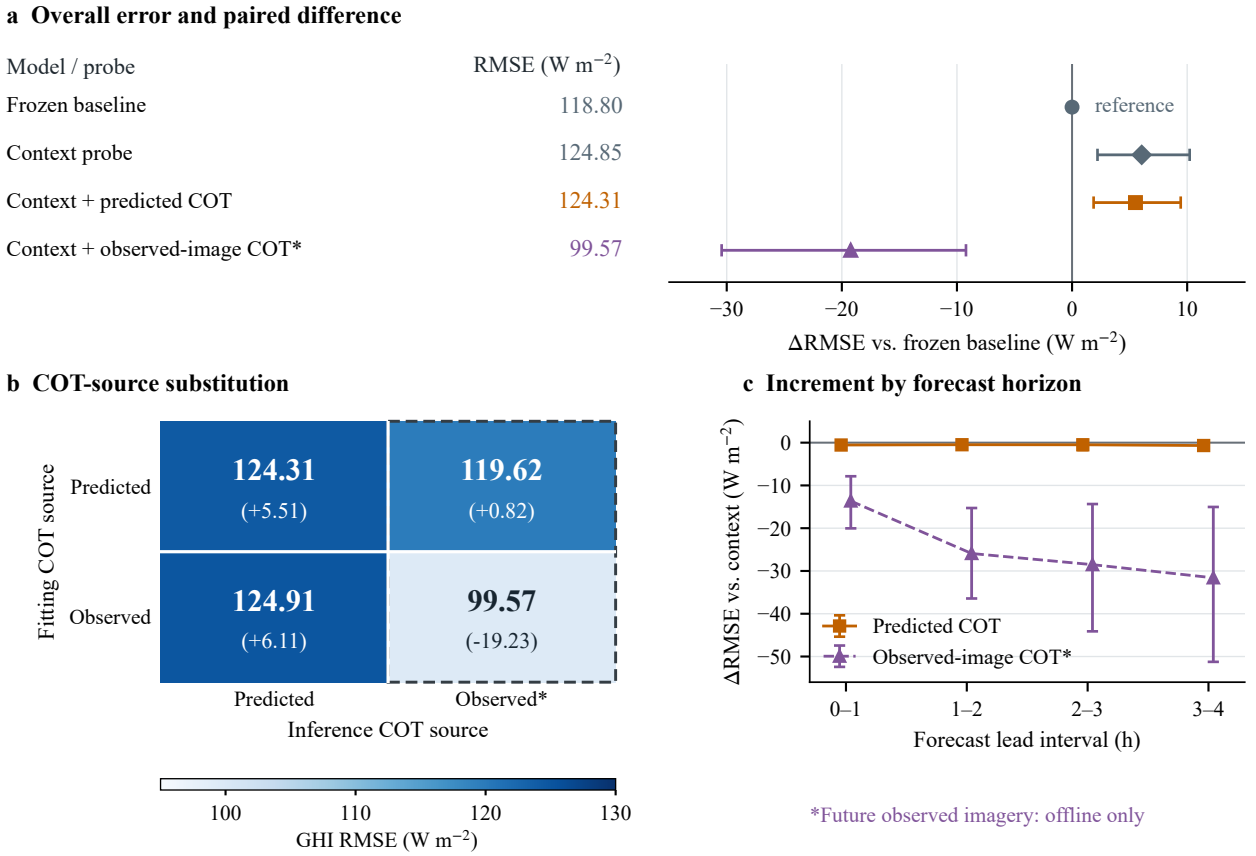


Figure S1: COT information under forecast and observed-image inputs. (a) GHI RMSE and paired differences from the frozen forecast baseline, which already uses historical COT. Context is a fixed linear residual probe using predicted AGRI, geometry, historical COT, lead and baseline output; the two augmented probes additionally receive future COT. (b) Reuse of the fitted probes under cross-substitution of the COT source. Entries are RMSE; parentheses give differences from the frozen baseline. The dashed inference column requires future observations. (c) Differences from the context probe by forecast horizon; 0–1 h includes +15 to +60 min. All panels use 32,838 common records from 50 development dates with date-block out-of-fold fitting. Scores average the station-specific RMSEs for three stations and three frozen-baseline sources equally. Whiskers are paired 95% percentile intervals from 5,000 target-date-cluster bootstrap resamples. Observed-image COT is a frozen retrieval from actual future AGRI, used offline; it is neither independent cloud truth nor a deployable input. Negative differences indicate lower error.

S3. Same-event trajectories at all three stations

Figure S2 retains an earlier forecast-source diagnostic initialized at 12:00 BJT on 6 August 2025. This was the earliest eligible correction-development date with all three stations, with initialization chosen nearest noon. It uses a historical-COT baseline and a fixed residual mapping, rather than the current synchronous models in Fig. 7. Gaps remain where a station lacks a valid 15-minute GHI target. The station curves describe one event, not three independent repetitions or an aggregate COT benefit.

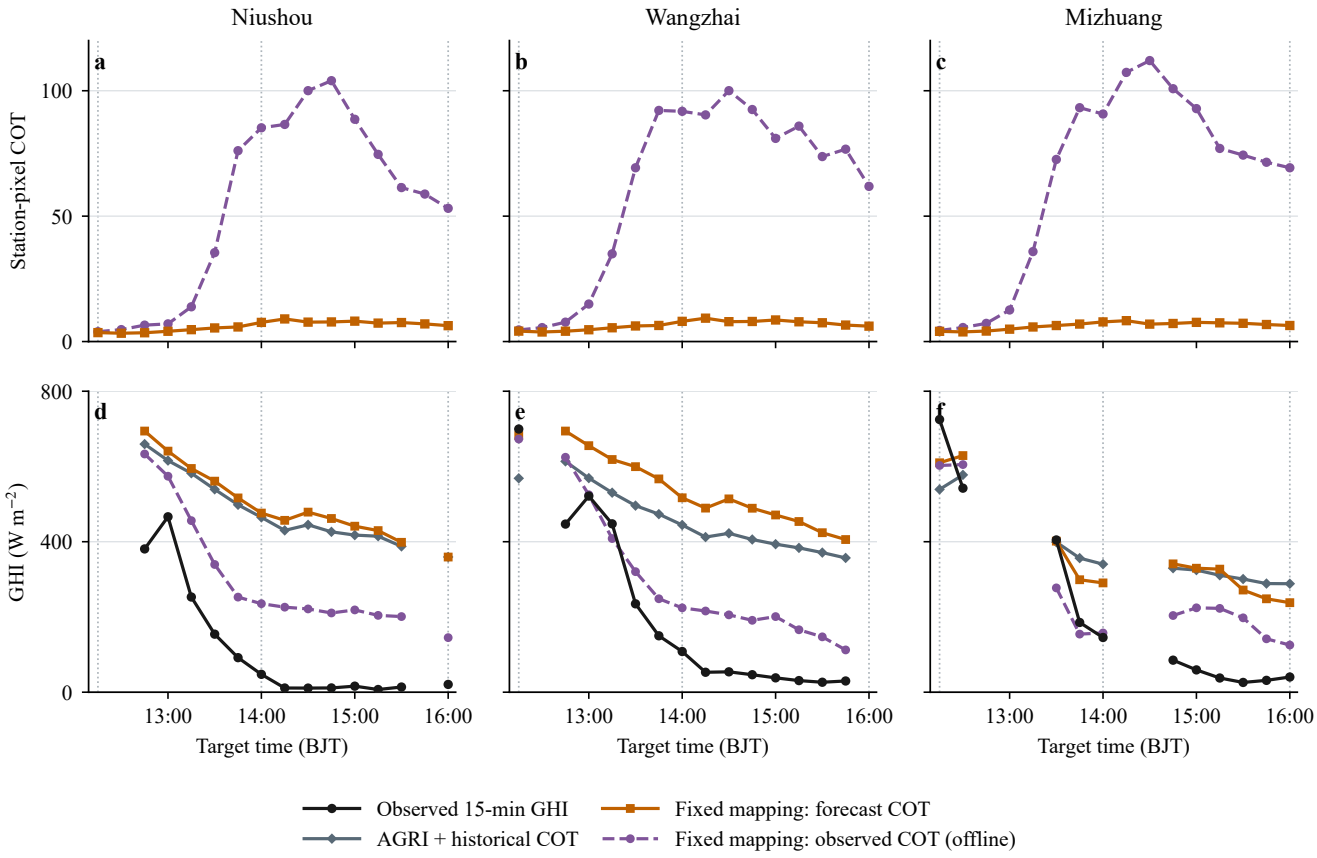


Figure S2: Station-pixel COT (a–c) and GHI (d–f) at Niushou, Wangzhai and Mizhuang in the same calendar-selected event. Purple dashed curves use observed future AGRI-derived COT offline; orange solid curves use SimVP-derived COT. The same reference-trained residual mapping is used for the two GHI source evaluations. Gray is the existing AGRI-plus-historical-COT baseline, not a no-COT control. The stations have 13, 14 and 11 scored GHI intervals, respectively; missing intervals remain gaps. Within-row scales are shared. Displayed model curves average the three frozen-baseline sources of this historical diagnostic; actual future AGRI-derived COT is used only offline.

S4. Estimation comparisons under the original solar-geometry preprocessing

These comparisons use stored models fitted with the original solar-geometry preprocessing. The Hubei CPP supervision is correctly matched in UTC, but both its retrieval inputs and GHI geometry retain that original preprocessing. The Hunan single-image comparison also retains its original geometry and fixed selected weights, with a common station-day quality exclusion. These validation estimates are reproducible configuration-specific results, not the primary verified-geometry estimation scores. The thickness-stratified Hubei results have the same provenance as its overall comparison.

Under the original geometry, adding COT lowers single-image GHI errors in both regional validation sets (Fig. S3). In Hubei, the station- and seed-averaged RMSE decreases from 77.285 to 69.142 $\mathrm{W\,m^{-2}}$, a reduction of 10.54%. The paired difference is -8.143 $\mathrm{W\,m^{-2}}$ (95% two-day-block interval: $[-14.999, -3.361]$). MAE decreases from 56.616 to 46.723 $\mathrm{W\,m^{-2}}$, and positive bias decreases from 24.799 to 13.236 $\mathrm{W\,m^{-2}}$. The evaluation contains 2,195 station–time records in ten two-day blocks, with three random initializations. All three station means improve: Niushou, 76.851 to 63.978; Wangzhai, 70.539 to 63.137; and Mizhuang, 84.466 to 80.312 $\mathrm{W\,m^{-2}}$.

The complementary Hunan single-image comparison contains 4,440 station–time records across 51 target dates. RMSE decreases from 108.181 to 103.995 $\mathrm{W\,m^{-2}}$ (3.87%), with a paired difference of -4.187 $\mathrm{W\,m^{-2}}$ and a 95% date-block interval of $[-6.652, -1.855]$. Sili improves from 103.558 to 99.877 $\mathrm{W\,m^{-2}}$ and Zhujia from 112.805 to 108.113 $\mathrm{W\,m^{-2}}$. The comparison is conditional on these preprocessing configurations; it does not isolate the effect of correcting solar geometry.

The Hunan representation controls provide context for this improvement (Table S3). Additional image features yield an RMSE of 108.014 $\mathrm{W\,m^{-2}}$, close to the baseline, whereas retrieval-model features yield 103.456 $\mathrm{W\,m^{-2}}$, comparable to explicit COT. Thus, cloud-supervised features can be useful without expressing that information solely as optical thickness. The result supports COT as a useful representation in this setting, but does not establish that explicit thickness is uniquely responsible for the entire gain. In the Hubei cloud-probability comparison, continuous cloud probability alone yields 74.053 $\mathrm{W\,m^{-2}}$; combining probability with COT yields 68.815 $\mathrm{W\,m^{-2}}$. Its additional reduction relative to COT alone is 0.47%, with a 95% interval of $[-1.23, 2.62]$%, providing no clear overall incremental benefit.

Table S3: Hunan single-image representation controls on the same 4,440 validation records. RMSE is averaged equally across two stations and three initializations.

Input representation	RMSE ($\mathrm{W\,m^{-2}}$)	Reduction (%)
AGRI + solar geometry	108.181	—
Additional image features	108.014	0.15
Explicit COT	103.995	3.87
Retrieval-model features	103.456	4.37

The Hubei reference-covered subsets show a marked dependence on optical thickness (Fig. S4). For near-zero, thin, moderate, and thick reference COT, RMSE decreases from 88.704 to 85.145, from 116.483 to 106.130, from 77.587 to 75.281, and from 57.139 to 34.527 $\mathrm{W\,m^{-2}}$, respectively. The corresponding reductions are 4.01%, 8.89%, 2.97%, and 39.57%. The thick-cloud subset contains 173 station–time records in four two-day blocks; its 95% reduction interval is $[28.04, 50.05]$%. Near-zero, thin, and moderate subsets contain 268, 71, and 211 records across six, three, and five blocks, respectively, and their reduction intervals include zero. The overall GHI result includes records without CPP and therefore is not a weighted average of these four subsets.

In the thick-cloud subset, mean bias decreases from +36.097 to +6.093 $\mathrm{W\,m^{-2}}$, consistent with correcting excessive transmitted irradiance under strong attenuation. This does not imply a universal downward correction: moderate-cloud bias changes from +14.490 to -11.023 $\mathrm{W\,m^{-2}}$, while near-zero bias increases from +6.218 to +14.284 $\mathrm{W\,m^{-2}}$. At Mizhuang, near-zero-COT RMSE increases from 114.775 to 118.948 $\mathrm{W\,m^{-2}}$. The strongest aggregate improvement therefore coexists with adverse responses in particular conditions and stations. The non-monotonic reductions and limited reference coverage preclude a general claim that benefit increases continuously with thickness.

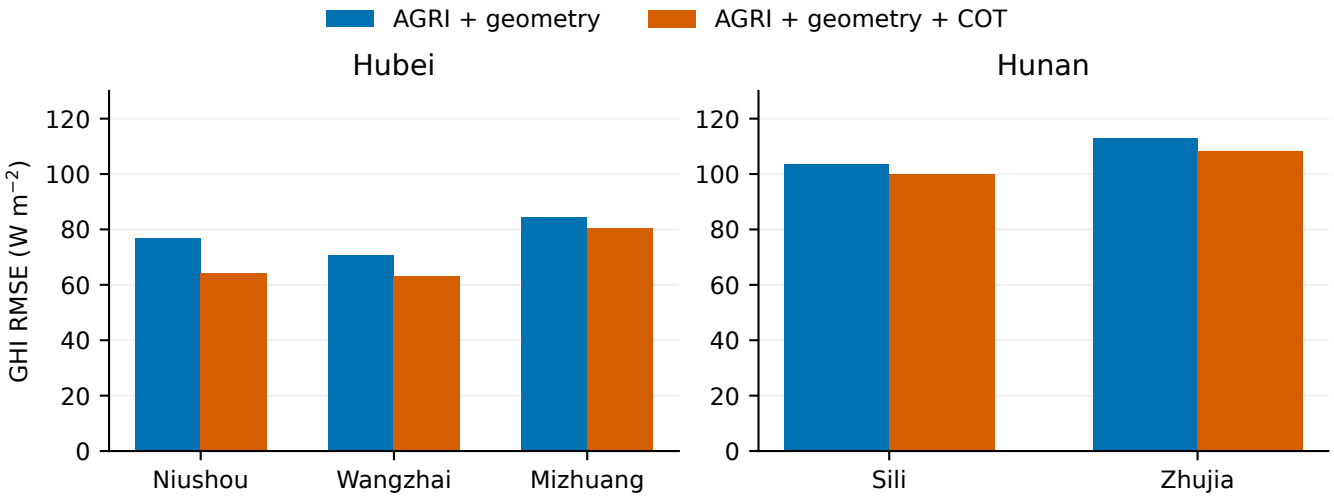


Figure S3: Station-wise GHI estimation errors using observed imagery, averaged over three initializations. Hubei uses 2,195 records; Hunan uses 4,440. The regional samples and models are independent.

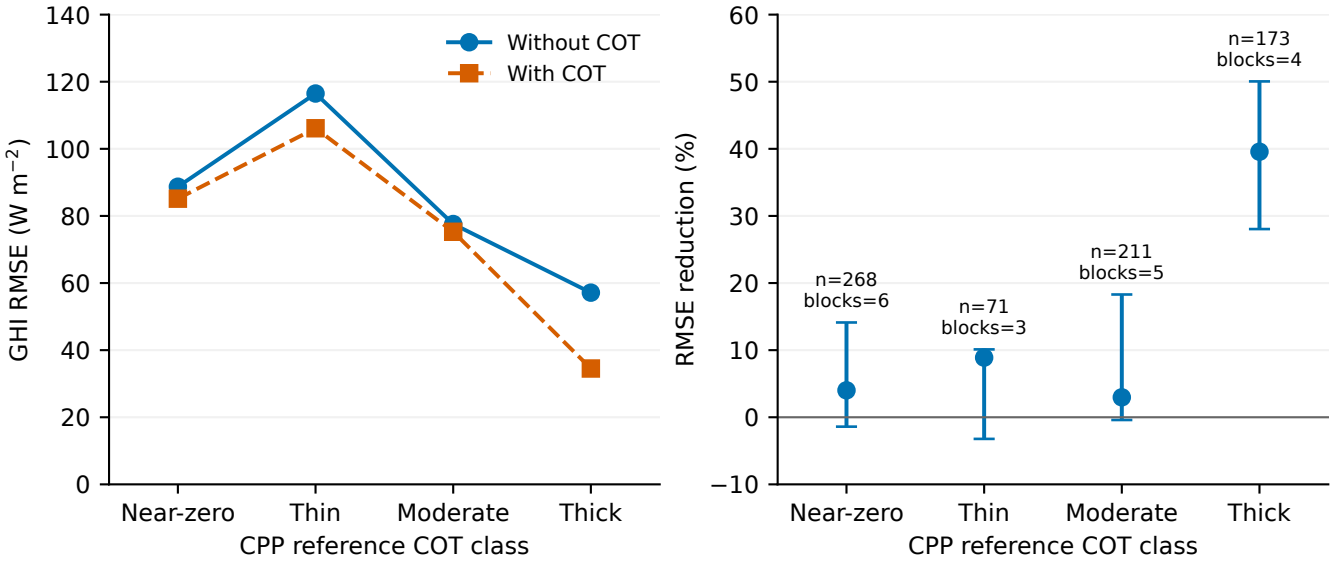


Figure S4: Hubei GHI errors (left) and relative RMSE reductions after adding COT (right), conditional on station-centre CPP COT. Error bars show 95% paired two-day-block intervals. Counts denote station–time records and date blocks, not independent pixels.

S5. Hubei forecast-trained comparisons and controlled input substitutions

The forecast-trained comparison separates historical-COT information from the conditional contribution of future COT. An imagery-only baseline and an otherwise matched historical-COT baseline are fitted on base-training dates; the latter receives eight observed-history COT fields. The historical-COT model is then frozen. Two equal-capacity gated residual heads are fitted on separate correction-training dates: the control has neutral-zero future-COT slots, while the augmented head receives COT from forecast AGRI. Both use the same examples, geometry, sample order and budget. Corrections are signed, with zero-initialized output layers, and the final clear-sky index is nonnegative. Base and correction training use 30 epochs, batch 1,536 and AdamW (learning rate 10^{-3}, weight decay 10^{-4}, cosine decay to 10^{-5}). The three random seeds are 42, 43 and 44. This is a separately trained forecasting comparison, not a direct reuse of the synchronous head's weights. Table S2 summarizes the four input definitions.

The forecast-trained comparison uses 15,275 common station–initialization–target records from 22 selected validation target dates between 22 August 2025 and 29 June 2026. Table S4 separates the historical-COT baseline from the two subsequent correction heads. Historical COT lowers equal-station, equal-seed RMSE from 105.595 to 103.670 W m^{-2}, a point reduction of 1.823%. The paired difference is -1.925 W m^{-2} with a 95% target-date block interval of $[-4.734, +0.670]$, so the apparent improvement is

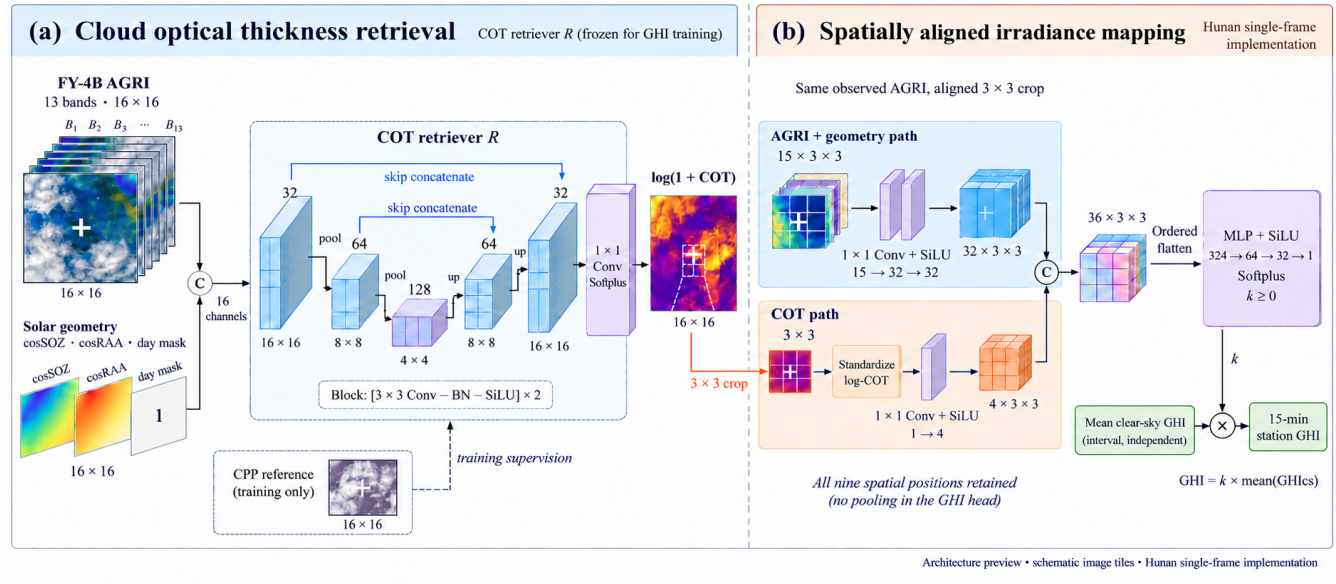


Figure S5: Single-frame COT-assisted GHI estimation, illustrated for the Hunan implementation. A 16×16 AGRI–geometry context enters a U-Net retriever; its nonnegative log-COT output is converted to physical COT. The GHI model reads aligned 3×3 AGRI, geometry and standardized log-COT fields. Ordered features are concatenated and flattened, then mapped to the clear-sky index and multiplied by interval-mean clear-sky GHI. Dashed paths denote supervision. Cloud textures are schematic, not observations. This AI-assisted visual preview depicts the method; regional model sizes and the Hubei auxiliary-slot variant are specified in Table S1.

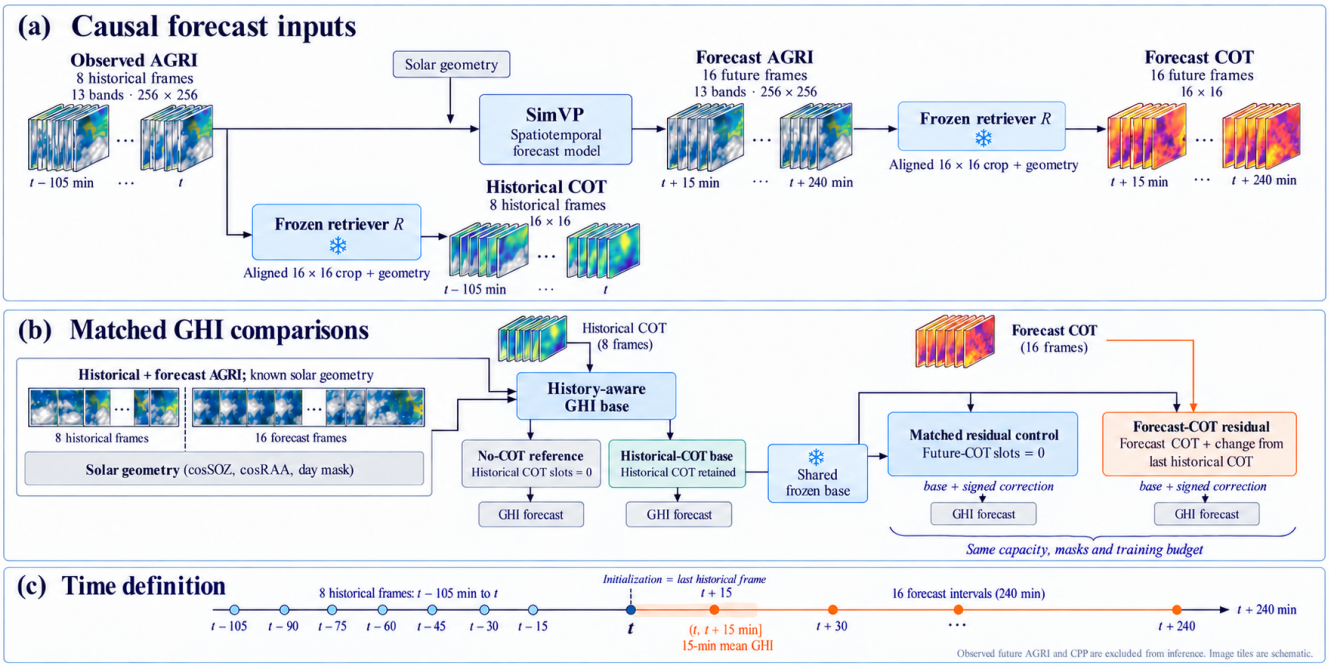


Figure S6: Forecast comparison and causal input flow. Eight observed historical AGRI frames supply the image predictor and historical COT. SimVP generates 16 future AGRI frames, from which the fixed retriever derives future COT. The four comparisons separate no COT, historical COT, a matched correction control, and a forecast-COT correction. The latter two share a frozen historical-COT base; neutral future-COT slots in the control preserve correction capacity. Signed corrections are zero-initialized and gated; unavailable future COT falls back to the base. Future observed imagery and CPP are excluded from this inference flow. Tensor drawings are illustrative, not measurements; input definitions are listed in Table S2. The GHI image branch uses 13 AGRI channels and two geometry channels; the daytime indicator in the shared geometry strip is used by retrieval and validity masking, rather than as an additional GHI image channel.

not resolved from zero by this analysis.

Adding forecast COT to the correction head changes RMSE from 103.783 to 103.808 W m^{-2}. Its difference from the matched-capacity control is +0.026 W m^{-2}, with interval $[-0.003, +0.059]$. Figure S7 retains the complete 16-lead comparison and distinguishes seed variation from date-block uncertainty. This comparison does not demonstrate an additional benefit from future COT; it also does not establish equivalence or universal absence of value.

Table S4: Forecast GHI RMSE (W m^{-2}) on common validation rows. Each value averages three seed-specific scores; the final row also averages stations equally. The correction heads share the trained historical-COT base.

Station	No COT	History COT	History + control	History + forecast COT
Niushou	97.017	96.299	96.665	96.701
Wangzhai	106.183	102.892	103.165	103.215
Mizhuang	113.586	111.819	111.519	111.510
Equal-station mean	105.595	103.670	103.783	103.808

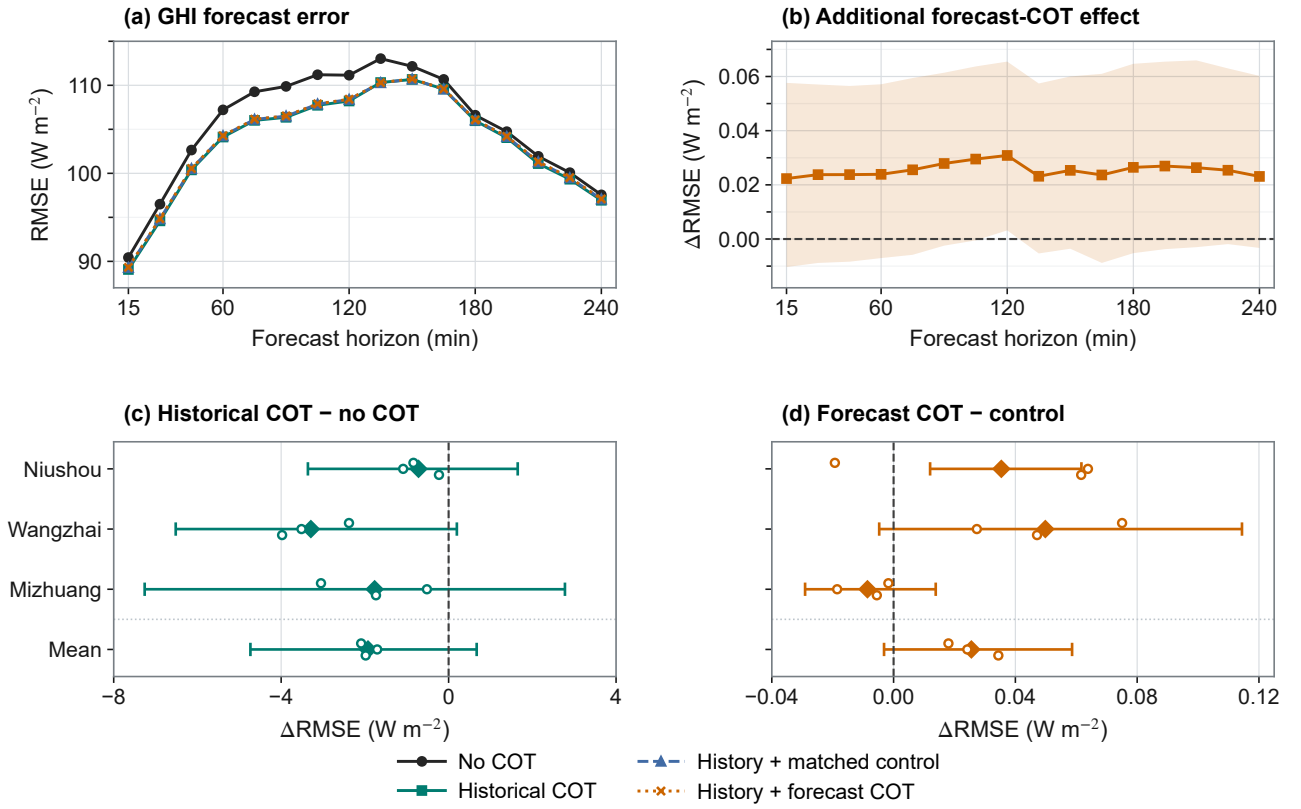


Figure S7: Paired GHI forecast evaluation on 15,275 common station–initialization–target records from 22 validation target dates. (a) Four model variants over +15–240 min from the last observed frame. (b) Forecast-COT correction minus the matched correction control, with pointwise 95% paired target-date bootstrap intervals. (c,d) Historical-COT-minus-no-COT and forecast-COT-minus-control differences for Niushou, Wangzhai, Mizhuang and their equal-station mean; both panels share this station order. Diamonds average three seed-specific scores; open circles show individual seeds; whiskers are 95% intervals from 5,000 paired date-block draws. Negative differences improve RMSE. Panel scales differ. The correction models share the frozen historical-COT base. Future observed AGRI/CPP is not an inference input. These are development-validation results, not an independent test.

A separate fixed-head diagnosis uses 8,111 common replacement rows from 12 development target dates. The control in this diagnosis already contains historical COT. Replacing forecast AGRI by observed target-time AGRI lowers its RMSE from 103.469 to 89.441 W m^{-2} (difference −14.028, interval $[-21.911, -6.053]$). However, adding source-matched future COT gives a difference of +0.025 under forecast AGRI and +0.140 under observed AGRI; both intervals include zero. Thus better image information is relevant to this mapping, but coherent observed image and COT inputs do not themselves recover an incremental COT benefit in the fixed head.

Table S5: MSE sensitivity decomposition on the same 8,111 replacement rows. O denotes observed-image inputs and F forecast-image inputs. Units are $W^2\ m^{-4}$. Points satisfy Eq. 10; interval limits are not additive.

Contrast	Point	Paired 95% interval
$E_{FO} - E_{OO}$	2778.1	[938.0, 5228.9]
$E_{OF} - E_{OO}$	−5.4	[−48.9, 33.5]
$E_{FF} - E_{FO} - E_{OF} + E_{OO}$	−51.6	[−109.2, −7.8]
$E_{FF} - E_{OO}$	2721.1	[920.4, 5105.2]

The image-source term is large in this conditional comparison, while the interaction is nonzero. These are sensitivities of a particular fixed mapping rather than an independent physical error budget. The replacement rows differ from the forecasting-validation cohort, so their absolute scores must not be used as a validation-to-oracle gain. Neither these contrasts nor an individual cloud event uniquely identifies displacement, optical growth or retrieval behavior as the cause.

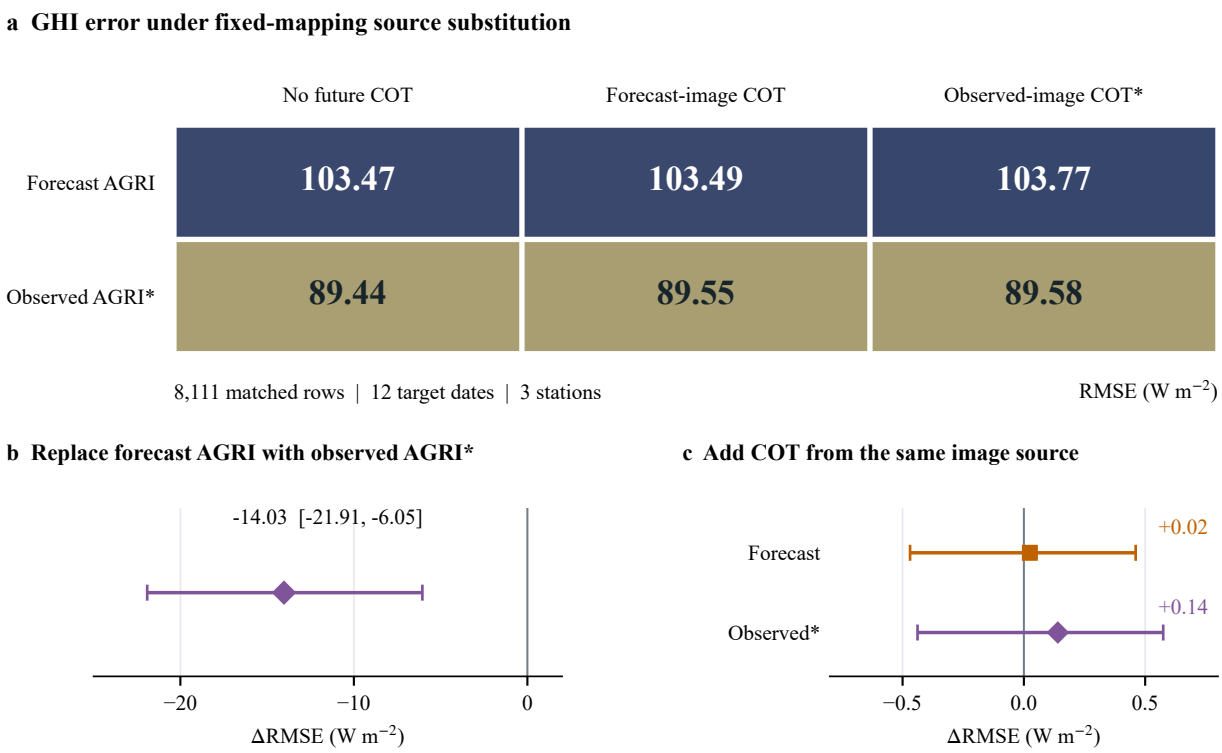


Figure S8: Sensitivity of a fixed irradiance mapping to image and COT sources. (a) Equal-station, equal-seed GHI RMSE on 8,111 common replacement rows from 12 development target dates (three stations, three seeds). All branches retain historical COT; "no future COT" removes only the additional future-COT correction input. (b) Replacing forecast AGRI with observed target-time AGRI in that control. (c) Adding source-matched future COT relative to the control with the same AGRI source. Intervals are 95% paired target-date block bootstrap intervals (5,000 draws); the horizontal scales in (b) and (c) differ. Observed future AGRI and its retrieved COT are offline diagnostic inputs. These comparisons do not constitute independent evaluation or deployable forecast improvements.

S6. Image-predictor comparison on common Hunan samples

The comparison retains 63,480 common validation records across 51 target dates for the two image predictors and the same observed-image-trained three-frame irradiance models. It is separate from the expanded regional DyDiff comparison in Table 4 and Fig. 8. Both methods show a negative point RMSE difference after adding predicted COT; this comparison does not establish that one image predictor has a larger COT benefit.

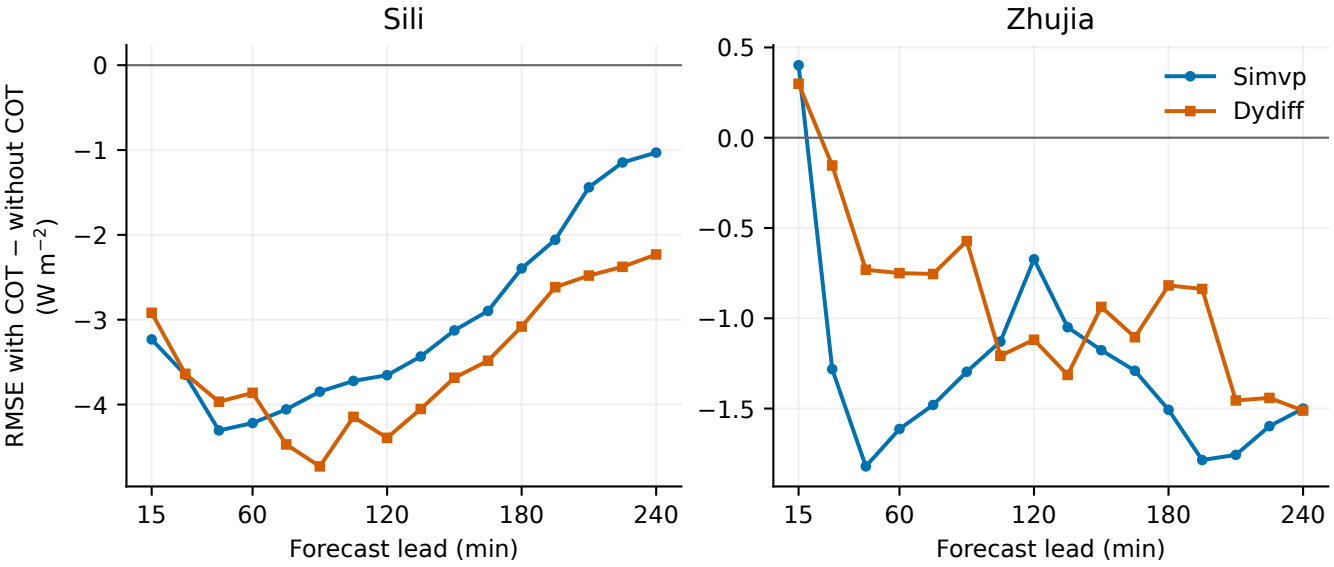


Figure S9: Lead-dependent change in Hunan GHI RMSE after adding predicted COT for two image predictors on the 63,480-record common validation cohort covering 51 target dates. Each point averages three random-seed-specific RMSE differences at the same station and lead; negative values favour COT. Curves are validation point estimates without lead-wise confidence intervals.

S7. Forecast cloud regimes and complete real-cloud cases

The reference-covered forecast analysis retains the original GHI scoring records and applies CPP classes only offline. Table S6 distinguishes forecast records, distinct station–target observations, and contiguous reference segments. Multiple origins do not create additional reference observations; segments do not count independent cloud systems.

Table S6: CPP coverage within the regional forecast validation samples. A multi-time segment contains at least two consecutive 15-minute thick-reference times.

Quantity	Hubei	Hunan
All GHI forecast records	24,451	66,086
CPP-valid forecast records	7,976	18,750
CPP-valid distinct station–targets	508	1,235
Thick-reference forecast records	2,141	2,435
Thick-reference distinct station–targets	139	165
Thick-reference target dates	6	25
Reference segments / multi-time segments	75 / 18	137 / 7

Table S7: Forecast GHI RMSE by reference COT class. Units are $\mathrm{W\,m^{-2}}$; differences are predicted-COT minus no-COT scores. Intervals use 5,000 paired target-date draws. N/D counts distinct station–targets/dates.

Region	Class	No COT	+ COT	Δ [95% interval]	N/D
Hubei	Near-zero	125.562	114.917	−10.645 [-12.479, -6.263]	216/8
Hubei	Thin	167.814	150.562	−17.252 [-26.761, -9.984]	24/3
Hubei	Moderate	86.965	94.089	7.125 [+4.320, +24.961]	129/5
Hubei	Thick	72.479	61.281	−11.199 [-25.927, +1.089]	139/6
Hunan	Near-zero	116.959	116.627	−0.332 [-3.898, +2.337]	449/29
Hunan	Thin	180.939	178.056	−2.883 [-5.657, -0.049]	176/28
Hunan	Moderate	149.587	148.408	−1.179 [-3.777, +1.085]	445/35
Hunan	Thick	92.134	89.464	−2.670 [-5.821, +0.690]	165/25

As a sensitivity analysis, averaging squared errors over origins before scoring each distinct station–target gives thick-subset RMSEs of 72.559 versus 61.234 $\mathrm{W\,m^{-2}}$ in Hubei and 90.838 versus 87.570 $\mathrm{W\,m^{-2}}$ in Hunan. Leaving out one date at a time gives paired differences from −14.616 to −6.830 and from −3.720 to −2.147 $\mathrm{W\,m^{-2}}$, respectively. These influence checks do not replace the main intervals that include zero.

For forecast COT itself, thick-cloud probability of detection, false-alarm ratio and critical success index are 0.717, 0.349 and 0.518 in Hubei, and 0.446, 0.365 and 0.355 in Hunan, at the same threshold of 30. Mean COT bias on thick-reference instances is −10.997 and −15.280, respectively. These dimensionless scores pool CPP-valid forecast instances rather than using the station–seed averaging of the GHI measures. Different regional reference coverage precludes interpreting them as a controlled regional model ranking.

The four cases below retain the earliest eligible improving and degrading multi-time segment per region. Each complete figure shows observed and forecast AGRI, CPP, COT retrieved from observed AGRI, COT retrieved from forecast AGRI, and aligned COT/GHI time series. Spatial arrays and processing match Fig. 10; missing observations remain missing. The Hunan forecast panels are explicitly magnified native 3×3 archived crops. The case scores below concern only the displayed initialization and seed 42, not all origins or the regional estimator.

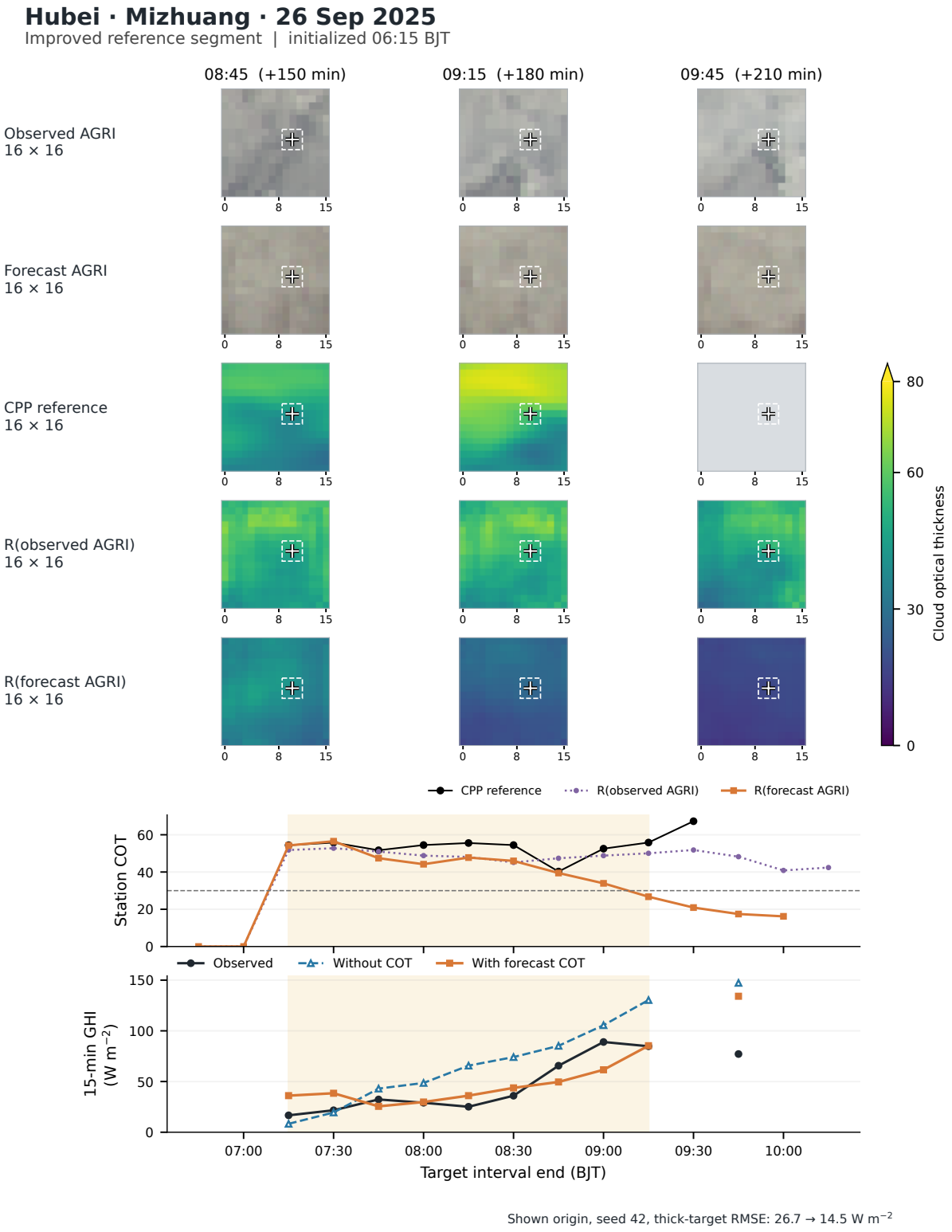


Figure S10: Mizhuang, Hubei, 2025-09-26; initialized at 06:15 BJT. The selected thick-reference segment is 07:15–09:15. Displayed-seed segment RMSE is 26.67 without COT and 14.51 W m^{-2} with predicted COT. The future observed fields are offline references, not forecast inputs.

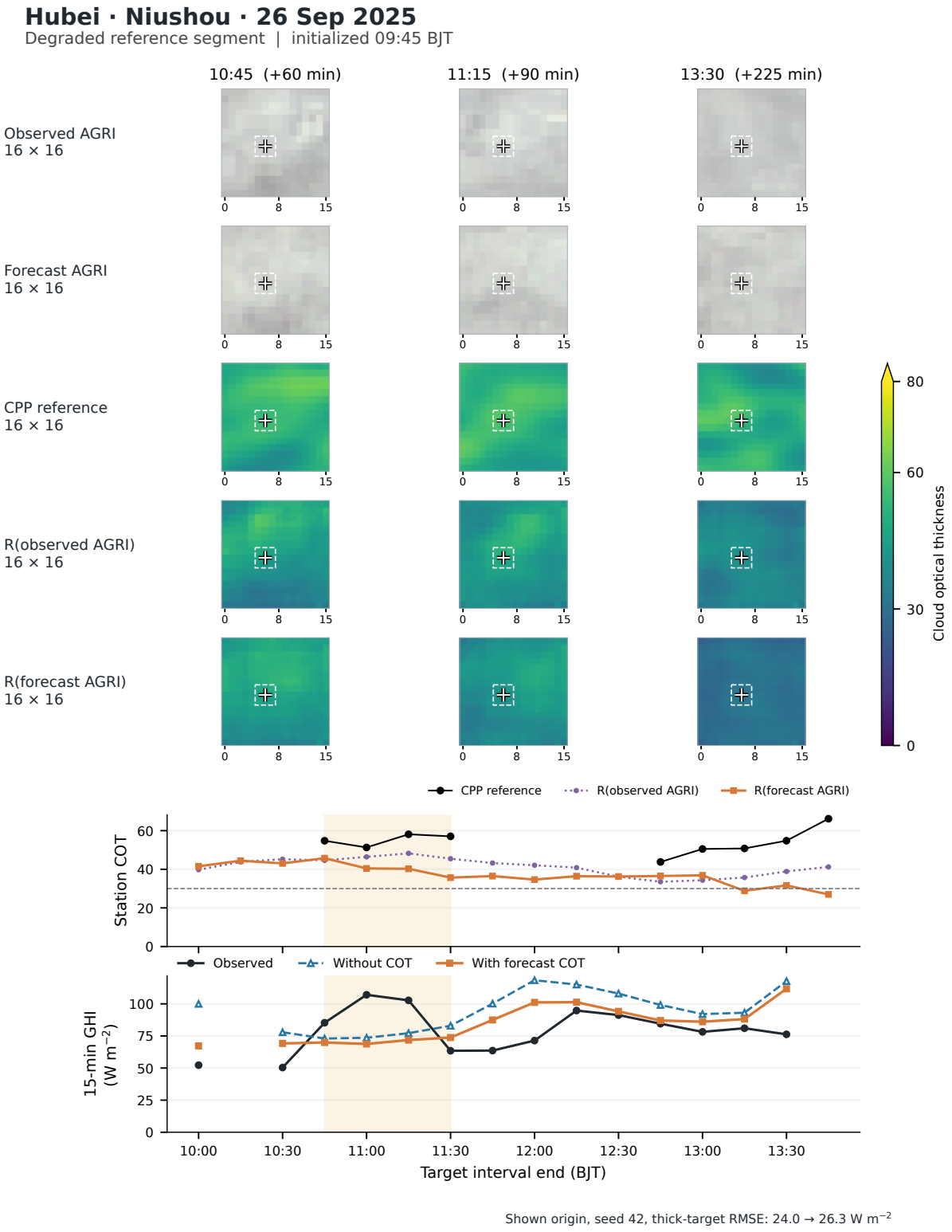


Figure S11: Niushou, Hubei, 2025-09-26; initialized at 09:45 BJT. The selected thick-reference segment is 10:45–11:30. Displayed-seed segment RMSE is 24.01 without COT and 26.29 W m^{-2} with predicted COT. The future observed fields are offline references, not forecast inputs.

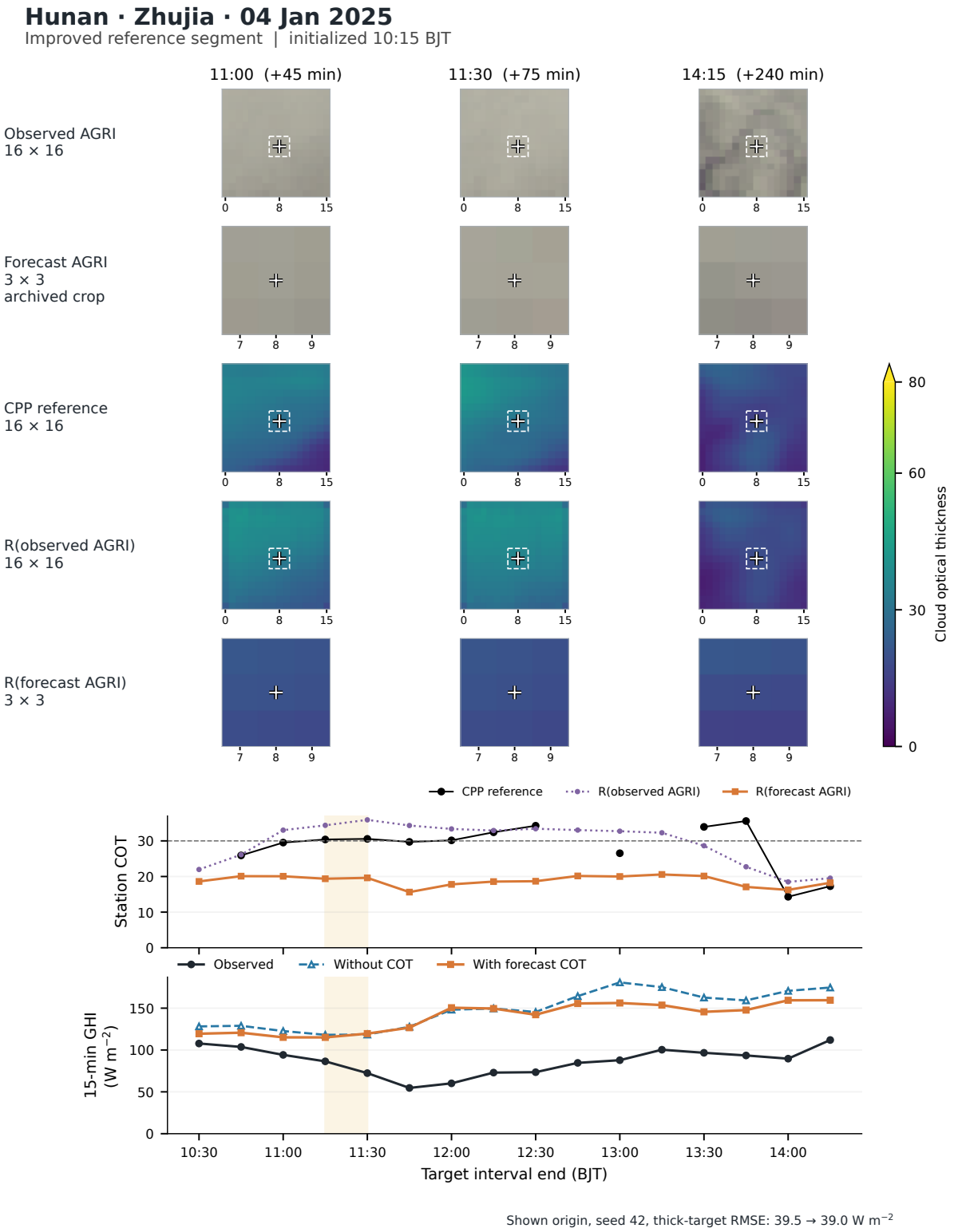


Figure S12: Zhujia, Hunan, 2025-01-04; initialized at 10:15 BJT. The selected thick-reference segment is 11:15–11:30. Displayed-seed segment RMSE is 39.54 without COT and 39.04 $\mathrm{W\,m^{-2}}$ with predicted COT. The future observed fields are offline references, not forecast inputs.

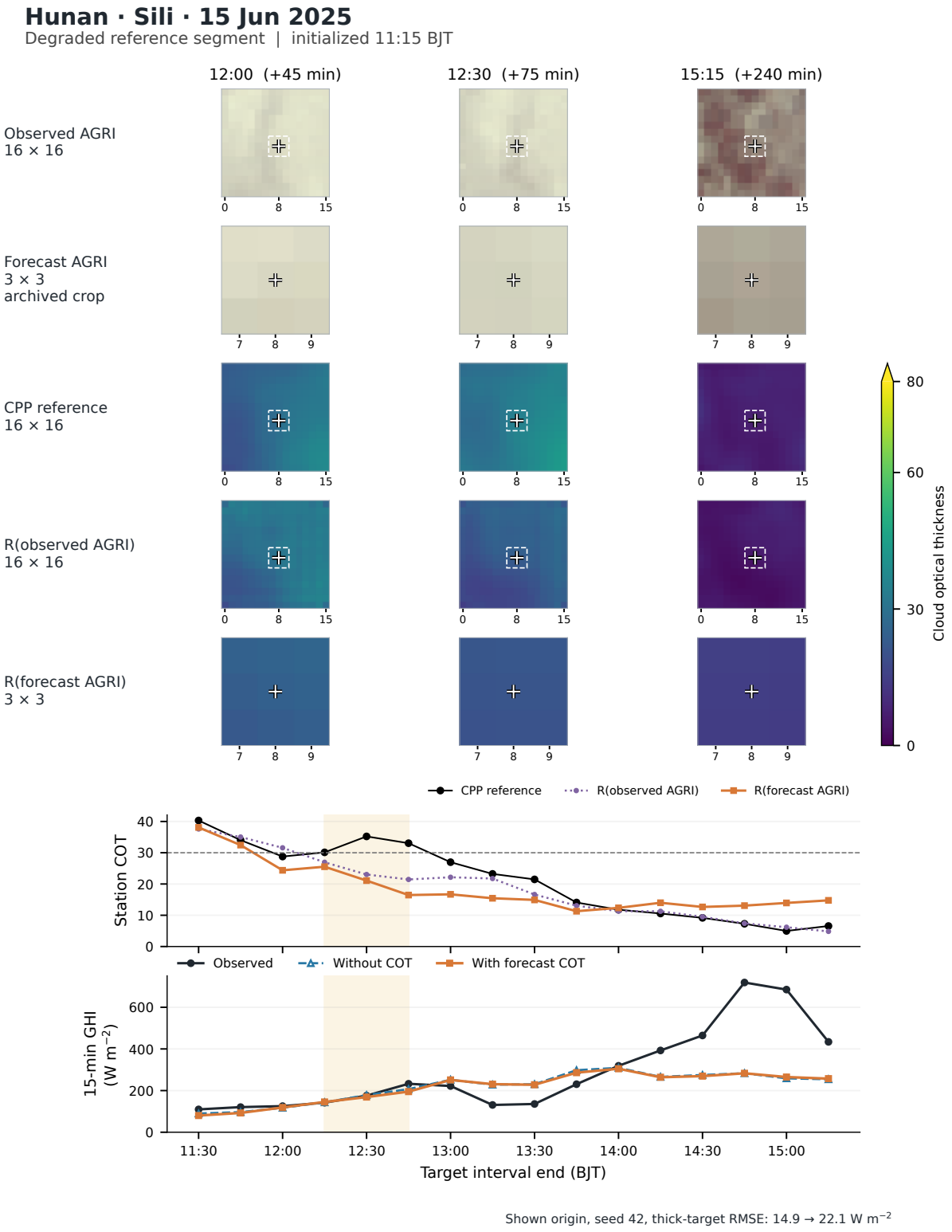


Figure S13: Sili, Hunan, 2025-06-15; initialized at 11:15 BJT. The selected thick-reference segment is 12:15–12:45. Displayed-seed segment RMSE is 14.87 without COT and 22.10 W m^{-2} with predicted COT. The future observed fields are offline references, not forecast inputs.